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Stream RAG: Instant and Accurate Spoken Dialogue Systems with Streaming Tool Usage

Siddhant Arora[1,2,*], **Haidar Khan**[1], **Kai Sun**[1], **Xin Luna Dong**[1], **Sajal Choudhary**[1], **Seungwhan Moon**[1], **Xinyuan Zhang**[1], **Adithya Sagar**[1], **Surya Teja Appini**[1], **Kaushik Patnaik**[1], **Sanat Sharma**[1], **Shinji Watanabe**[2], **Anuj Kumar**[1], **Ahmed Aly**[1], **Yue Liu**[1], **Florian Metze**[1], **Zhaojiang Lin**[1]

[1]Meta, [2]Carnegie Mellon University
[*]Work done at Meta

End-to-end speech-in speech-out dialogue systems are emerging as a powerful alternative to traditional ASR–LLM–TTS pipelines, generating more natural, expressive responses with significantly lower latency. However, these systems remain prone to hallucinations due to limited factual grounding. While text-based dialogue systems address this challenge by integrating tools such as web search and knowledge graph APIs, we introduce the first approach to extend tool use directly into speech-in speech-out systems. A key challenge is that tool integration substantially increases response latency, disrupting conversational flow. To mitigate this, we propose Streaming Retrieval-Augmented Generation (*Streaming RAG*), a novel framework that reduces user-perceived latency by predicting tool queries in parallel with user speech, even before the user finishes speaking. Specifically, we develop a post-training pipeline that teaches the model when to *issue tool calls* during ongoing speech and how to generate spoken summaries that fuse audio queries with retrieved text results, thereby improving both *accuracy and responsiveness*. To evaluate our approach, we construct AudioCRAG, a benchmark created by converting queries from the publicly available CRAG dataset into speech form. Experimental results demonstrate that our *streaming RAG* approach increases QA accuracy by up to 200% relative (from 11.1% to 34.2% absolute) and further enhances user experience by reducing tool use latency by 20%. Importantly, our *streaming RAG* approach is modality-agnostic and can be applied equally to typed input, paving the way for more agentic, real-time AI assistants.

Date: October 3, 2025
Correspondence: siddhana@andrew.cmu.edu, zhaojiang@meta.com
Meta

1 Introduction

Spoken Dialogue Systems (SDS) are foundational to many everyday technologies, powering intelligent assistants such as Alexa and Siri, as well as interactive voice response systems in customer service. With the rapid expansion of SDS capabilities to mobile phones and wearable devices, the need for robust, scalable, and generalizable solutions has never been greater. Traditionally, SDS have relied on cascaded pipelines composed of multiple modules—including voice activity detection (VAD), automatic speech recognition (ASR), natural language understanding (NLU), natural language generation (NLG), and text-to-speech (TTS) synthesis—each introducing potential points of failure and latency (Glass, 1999; Huang et al., 2024). Recently, end-to-end (E2E) SDS (Xie and Wu, 2024; Nguyen et al., 2023; Meng et al., 2024; Zhang et al., 2024; Arora et al., 2025b) have been proposed, which directly generate spoken responses from speech input within a unified architecture. This E2E approach not only mitigates error propagation across modules but also captures non-phonemic information more effectively, resulting in significantly lower inference time and computational overhead, and paving the way for more natural and efficient conversational experiences.

Despite these advances, current E2E SDS are fundamentally constrained by their reliance on internalized knowledge from static training data, which often results in responses that lack factual grounding or fail to reflect the most up to date information. This shortcoming is particularly critical for action-oriented or knowledge-seeking tasks, such as booking hotels or answering questions about current events. In contrast, text-based conversational assistants have begun to overcome these limitations by integrating external tools

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串流 RAG：具備串流工具使用的即時且精準的語音對話系統

Siddhant Arora[1],2,[*], **海達爾・汗1, 孫凱1, 董欣蘭1, 查達里・查達里1, 文承萬1, 張欣源1, 阿迪提亞・薩加爾1, 貝雅・阿皮尼1, 帕特納克・考希克1, 沙納特・夏爾馬1, 渡邊新治2, 安努・庫瑪1, 阿米德・阿利1, 劉岳1, 弗羅里安・梅澤1, 林兆江**[1]

[1]Meta, [2]卡內基梅隆大學
[*]Meta 的研究工作

端到端的語音輸入語音輸出對話系統正作為傳統 ASR–LLM–TTS 管線的強大替代方案而興起，能夠產生更自然、更具表達力的回應，且延遲顯著降低。然而，這些系統仍容易產生虛構內容，因為事實基礎有限。雖然基於文本的對話系統透過整合網路搜尋和知識圖形 API 等工具來解決這個挑戰，但我們介紹了第一種將工具使用直接擴展到語音輸入語音輸出系統的方法。一個關鍵挑戰在於，工具整合會顯著增加回應延遲，打亂對話流程。為了緩解這個問題，我們提出串流檢索增強生成（Streaming RAG），一個新框架，透過在用戶發言期間並行預測工具查詢，甚至在使用者完成發言之前，來降低使用者感知的延遲。具體來說，我們開發了一個後訓練流程，教導模型在持續語音期間何時發出工具呼叫，以及如何生成融合音頻查詢與檢索文本結果的語音摘要，從而提高準確性和反應速度。為了評估我們的方法，我們構建了 AudioCRAG，一個基準，透過將公開可用的 CRAG 資料集的查詢轉換為語音形式來創建。實驗結果顯示，我們的串流 RAG 方法相對提高了 QA 準確性達到 200%（從 11.1% 提高到 34.2% 的絕對值），並進一步透過降低工具使用延遲 20% 來提升用戶體驗。重要的是，我們的串流 RAG 方法是模態無關的，可以同樣應用於鍵盤輸入，為更具自主性、實時的 AI 助手鋪平道路。

日期：2025年10月3日 通信：siddhana@andrew.cmu.edu, zhaojiang@meta.com
Meta

1 引言

語音對話系統（SDS）是許多日常技術的基礎，為Alexa和Siri等智能助理以及客戶服務中的互動語音響應系統提供支持。隨著SDS功能快速擴展到手機和可穿戴設備，對於穩固、可擴展和普適性解決方案的需求從未如此迫切。傳統上，SDS依賴由多個模塊組成的級聯管道，包括語音活動檢測（VAD）、自動語音識別（ASR）、自然語言理解（NLU）、自然語言生成（NLG）和文本到語音（TTS）合成，每個模塊都引入潛在的故障點和延遲（Glass，1999；Huang等人，2024）。最近，已提出端到端（E2E）SDS（Xie和Wu，2024；Nguyen等人，2023；Meng等人，2024；Zhang等人，2024；Arora等人，2025b），它們在統一的架構中直接從語音輸入生成語音響應。這種E2E方法不僅減少了跨模塊的錯誤傳播，還更有效地捕捉非音素信息，結果顯著降低了推理時間和計算開銷，為更自然和高效的對話體驗鋪平了道路。

儘管有這些進步，目前的端到端學習系統（E2E SDS）基本上受到其依賴靜態訓練數據所內化的知識的限制，這往往導致回應缺乏事實依據或無法反映最新信息。這項缺點對於行動導向或知識尋求任務來說尤其關鍵，例如預訂酒店或回答有關當前事件的問題。相比之下，基於文本的對話助理已經開始通過整合外部工具來克服這些限制。

through Retrieval-Augmented Generation (RAG)(Yang et al., 2024; Chen et al., 2023a,c; Gao et al., 2024a,b), dynamically retrieving relevant information from sources like web search, knowledge graphs (KG), and real-time APIs. Yet, the integration of such tool use into E2E SDS remains largely unexplored. A key challenge is that while external tools can substantially improve factual accuracy, invoking them often introduces additional latency, leading to awkward silences that disrupt the natural conversational flow. This raises a research question: *How can we trade-off between accuracy and responsiveness for developing SDS that feel both intelligent and natural?*

In this paper, we present, to the best of our knowledge, the first *speech-in, speech-out language model that seamlessly integrates external tool invocation with low latency.* The key idea is a **Streaming RAG** strategy that generates tool queries in parallel with user speech, often times even before the user has finished speaking (Fig. 1). A naive implementation of streaming queries, however, faces two challenges: (1) queries issued from partial speech may be suboptimal, yielding distracting tool outputs and inaccurate responses; and (2) unnecessary tool calls may be triggered, wasting computational resources. We introduce effective modeling techniques to address these challenges and make the following contributions.

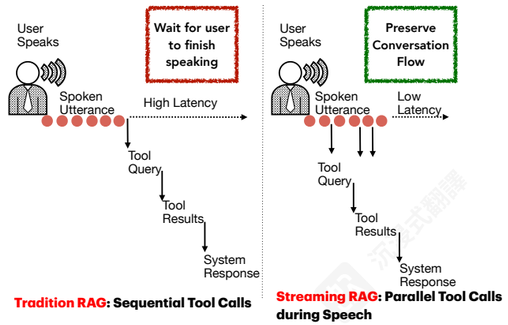


Figure 1 Comparison of Traditional RAG with proposed Streaming RAG which fires tool queries in parallel with user speech.

Contribution 1: We introduce a formal framework for tool integration in speech-in speech-out systems and empirically show that leveraging web search and KG APIs significantly enhances factual question answering. Evaluating three state-of-the-art (SOTA) models, Qwen-OMNI (Xu et al., 2025), OpusLM (Tian et al., 2025), and Kimi-Audio (Ding et al., 2025), we find that external tool integration delivers substantial performance gains, boosting accuracy by up to 140% relative (from 11.1% to 26.3% absolute). However, tool usage also introduces considerable latency, increasing first-token response time by 2.3x.

Contribution 2: To address this, we propose *Streaming Retrieval-Augmented Generation* (*Streaming RAG*), the *first* framework that empowers the system to trigger tool queries in parallel with user speech, even before the user finishes speaking. Within this framework, we introduce two novel approaches: (1) *Fixed-Interval Streaming RAG*, which issues tool queries at regular intervals during speech input and carefully examines quality of retrieval results on the full query to guarantee response quality, and can be incorporated into any speech-in, speech-out model without post-training; (2) *Model-Triggered Streaming RAG*, which post-trains the model to intelligently determine optimal query timing based on the evolving user utterance to save computation resources. Our results demonstrate that our proposed *Model-Triggered Streaming RAG* delivers over 200% relative improvement in accuracy (from 11.1% to 34.2% absolute in T. 1) compared to the no-tool baseline, while also reducing tool result generation latency by 20%. Though designed for speech-in speech-out systems, streaming RAG can also be adapted in cascaded SDS, or even chatbots as users type.

Contribution 3: Finally, we introduce AudioCRAG, a benchmark created by recording spoken queries from the CRAG (Yang et al., 2024) dataset, enabling robust evaluation of tool usage capabilities in speech-in speech-out systems. To support future research, we will open source our training code and AudioCRAG-Human benchmark, supporting future research in tool-integrated voice assistants.

2 Related studies

2.1 Benchmarks for Tool Usage

Recent advances in benchmarking text-based dialogue systems for tool usage (Chen et al., 2023b; Ouyang et al., 2025; Cheng and Dou, 2025; Cohen et al., 2025; Xiong et al., 2024a) have primarily focused on evaluating factual question answering and task completion within simulated environments (detailed related work discussion in

透過檢索增強生成 (RAG)(Yang et al., 2024; Chen et al., 2023a,c; Gao et al., 2024a,b)，動態地從網路搜尋、知識圖形 (KG) 和實時 API 等來源中檢索相關資訊。然而，將這種工具使用整合到端到端 (E2E) SDS 中仍然相當未被探討。一個主要的挑戰在於，雖然外部工具可以顯著提高事實準確性，但呼叫它們往往會引入額外的延遲，導致尷尬的沉默，打斷自然的對話流程。這提出了一個研究問題：我們如何能在準確性和反應速度之間取得平衡，以開發既智能又自然的 SDS？

在本研究中，根據我們所知，我們提出第一個語音輸入、語音輸出的語言模型，能無縫整合外部工具呼叫並具低延遲。關鍵概念是串流RAG策略，該策略會與使用者語音平行生成工具查詢，有時甚至在使用者尚未說完之前就生成（圖1）。然而，串流查詢的直觀實現面臨兩個挑戰：(1) 由部分語音發出的查詢可能不理想，導致干擾性的工具輸出和不準確的回應；(2) 可能觸發不必要的工具呼叫，浪費計算資源。我們引入有效的建模技術來解決這些挑戰，並做出以下貢獻。

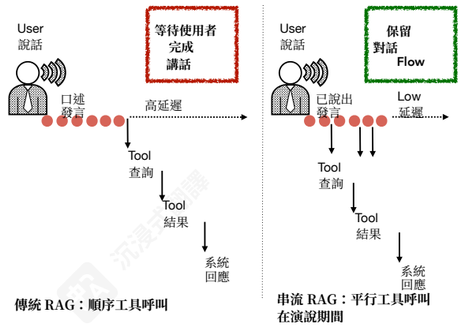


圖1 比較傳統RAG與所提出的串流RAG，串流RAG會與使用者語音平行發出工具查詢。

貢獻1：我們引入一個正式框架
用於語音輸入、語音輸出系統中的工具整合，並實證顯示利用網路搜尋和知識圖形API顯著提升事實性問答能力。評估三個最先進（SOTA）模型，Qwen-OMNI（Xu等人，2025年）、OpusLM（Tian等人，2025年）和Kimi-Audio（Ding等人，2025年），我們發現外部工具整合能顯著提升性能，相對準確率提升了140%（從11.1%提升到26.3%絕對值）。然而，工具使用也引入了顯著的延遲，將第一個Token的回應時間增加了2.3倍。

貢獻 2：為了解決這個問題，我們提出「串流檢索增強生成」（Streaming RAG），這是第一個能讓系統在用戶發言時甚至用戶尚未結束發言前，就能與用戶語音平行觸發工具查詢的框架。在此框架中，我們引入兩種新方法：(1)「固定間隔串流 RAG」，在語音輸入期間定期發出工具查詢，並仔細檢查完整查詢的檢索結果品質以保證回應品質，且無需後訓練即可融入任何語音輸入、語音輸出模型；(2)「模型觸發串流 RAG」，透過後訓練讓模型能根據用戶語句的演變智能判斷最佳查詢時機以節省計算資源。我們的結果顯示，我們提出的「模型觸發串流 RAG」相較於無工具基準線，準確率提升了 200% 以上（在 T. 1 中從 11.1% 提升至 34.2%），同時也將工具結果生成延遲減少了 20%。雖然設計用於語音輸入、語音輸出系統，但串流 RAG 也可以應用於串聯式 SDS，甚至用戶打字時的聊天機器人。

貢獻三：最後，我們介紹 AudioCRAG，這是一個由記錄 CRAG (Yang et al., 2024) 資料集中語音查詢所創建的基準，能夠在語音輸入語音輸出系統中進行穩健的工具使用能力評估。為了支持未來研究，我們將開源我們的訓練程式碼和 AudioCRAG-Human 基準，支持未來工具整合語音助手的相關研究。

2 相關研究

2.1 工具使用基準

近期針對工具使用進行基準測試的文本對話系統研究進展 (Chen 等人，2023b；Ouyang 等人，2025；Cheng 與 Dou，2025；Cohen 等人，2025；Xiong 等人，2024a) 主要集中在評估模擬環境中的事實性問答和任務完成 (相關工作討論詳見

Appendix A, B). The CRAG benchmark (Yang et al., 2024) is a leading example, featuring 4,409 question-answer pairs and providing mock APIs for both web and KG search. Recent benchmarks (Meta CRAG-MM Challenge Organizers, 2025; Ma et al., 2024; Jang et al., 2025) have extended tool-augmented dialogue evaluation to multimodal input and longer-context scenarios. While these benchmarks have significantly advanced the evaluation of tool-augmented dialogue systems, they remain largely limited to text-based outputs and do not fully address the unique challenges presented by speech-in speech-out systems.

2.2 E2E Spoken Dialogue Systems

Several E2E spoken dialogue systems (Xu et al., 2025; Xie and Wu, 2024; Arora et al., 2025a; Nguyen et al., 2023; Meng et al., 2024; Zhang et al., 2024; Arora et al., 2025b) have recently been introduced, demonstrating impressive semantic understanding and high audio quality in their responses. However, these systems have not yet been trained or evaluated for their ability to *use external tools*. Another research direction (Feng et al., 2025) explores E2E RAG for direct speech-to-text retrieval, utilizing multimodal embeddings for enabling speech utterances to directly retrieve relevant text. While this method outperforms models without RAG, it is primarily limited to *speech-to-text* scenarios. Its ability to access KGs and other APIs, critical for real-world applications, remains unproven. Furthermore, the retrieval scope is limited, as experiments are conducted with retrieval restricted to just 10 paragraphs, whereas modern RAG benchmarks require searching across thousands of web pages. Although recent studies (Maben et al., 2025) have developed web interfaces that integrate tools into speech-in *speech-out* scenarios using cascaded pipelines, comprehensive empirical investigations of E2E speech-in speech-out systems and, crucially, systematic analyses of user-perceived latency, remain largely unexplored. In this work, we address these gaps by developing a comprehensive framework for tool integration in E2E speech-in speech-out systems and designing benchmarks to quantitatively assess the tool usage capabilities of SOTA models. Furthermore, recognizing the importance of latency for natural conversational flow, we introduce novel streaming RAG methods that not only enhance tool usage performance but also reduce user-perceived latency.

3 Methodology

A RAG spoken conversation system takes an audio question Q as input and outputs a spoken answer A. Let the ASR transcript of audio question be X^{asr} and the ASR transcript of the spoken answer be X^{res}. Answers are generated by speech-in speech-out models, leveraging both the model's internal knowledge and information retrieved from external sources. To incorporate external information, the model needs to formulate a tool query Q^T to retrieve relevant results R from an external tool T.

3.1 Tool Integration for Speech-in Speech-Out LLMs

Figure 2 illustrates our proposed formulation for integrating external tools into speech-in speech-out systems. We introduce a two-stage inference approach: *Query Generation* and *Response Generation*. In the Query Generation stage, the system processes an audio question and generates queries for each external tool to retrieve relevant information by maximizing the posterior distribution $P(Q^T|Q)$ (Examples of generated queries are provided in T. 7 in the Appendix.). In the Response Generation stage, the retrieved results R from these tools are combined with the original audio question and input into the model to generate the final spoken response by maximizing the posterior distribution $P(A|Q,R)$. By conditioning the final output generation on the input audio, this formulation not only provides a simple and effective mechanism for

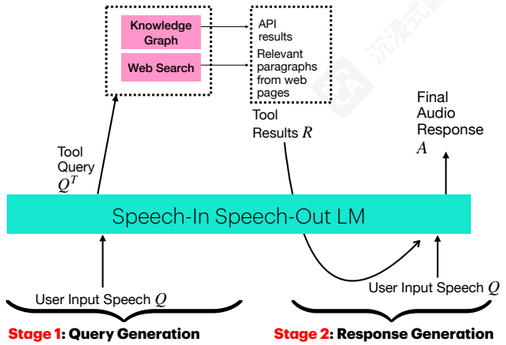


Figure 2 Proposed formulation for integrating tool usage in E2E speech-in, speech-out dialogue systems using a two-stage inference approach.

附錄A、B)。CRAG基準測試 (Yang 等人，2024) 是一個領先的範例，包含4,409個問答對，並提供網路和知識圖形搜索的模擬API。近期基準測試 (Meta CRAG-MM挑戰組織者，2025；Ma 等人，2024；Jang 等人，2025) 已將工具增強對話評估擴展到多模態輸入和長文本情境。雖然這些基準測試顯著推進了工具增強對話系統的評估，但它們仍然主要限於文本輸出，並未完全應對語音輸入語音輸出的系統所提出的獨特挑戰。

2.2 語音對話系統

近年來已推出數個語音對話系統（Xu et al., 2025; Xie and Wu, 2024; Arora et al., 2025a; Nguyen et al., 2023; Meng et al., 2024; Zhang et al., 2024; Arora et al., 2025b），在其回應中展現出令人印象深刻的語義理解能力與高音質。然而，這些系統尚未接受過或評估過其使用外部工具的能力。另一個研究方向（Feng et al., 2025）探討了直接語音到文字的對話式檢索，利用多模態嵌入使語音陳述能直接檢索相關文字。雖然此方法優於沒有 RAG 的模型，但它主要限於語音到文字的場景。其存取知識庫（KGs）和其他 API 的能力——對於實際應用至關重要——仍未被證實。此外，檢索範圍有限，因為實驗僅限制在檢索10段文字，而現代 RAG 基準則要求搜尋跨越數千頁網頁。儘管近年來的研究（Maben et al., 2025）已開發出整合工具至語音輸入語音輸出場景的網頁介面，使用串聯管線，但對語音輸入語音輸出系統的全面實證研究，以及至關重要的用戶感知延遲的系統分析，仍大體未被探討。在本工作中，我們透過開發一個全面的工具整合框架至語音輸入語音輸出系統，並設計基準來量化評估 SOTA 模型的工具使用能力，來填補這些差距。此外，認識到延遲對自然對話流程的重要性，我們引入了新的串流 RAG 方法，不僅增強了工具使用性能，也降低了用戶感知延遲。

3 方法論

一個 RAG 語音對話系統以音訊提問 Q 為輸入，並輸出語音答案 A。讓音訊提問的 ASR 說明文字為 X^{asr}，而語音答案的 ASR 說明文字為 X^{res}。答案由語音輸入語音輸出模型產生，利用模型內部知識以及從外部來源獲取的資訊。為了整合外部資訊，模型需要制定一個工具查詢 Q^T 以從外部工具 T 獲取相關結果 R。

3.1 語音輸入語音輸出 LLM 的工具整合

圖2說明了我們提出的整合外部工具至語音輸入-語音輸出系統的公式。我們引入一種兩階段推論方法：查詢生成與回應生成。在查詢生成階段，系統處理一個語音問題，並為每個外部工具生成查詢以據後驗分佈$P(Q^T|Q)$（生成的查詢範例提供於附錄T. 7中）。在回應生成階段，從這些工具中取得的結果 R 與原始語音問題結合，並輸入至模型，以據後驗分佈 $P(A|Q,R)$生成最終的語音回應。透過在輸入語音上條件化最終輸出生成，這個公式不僅提供了一個簡單且有效的機制以

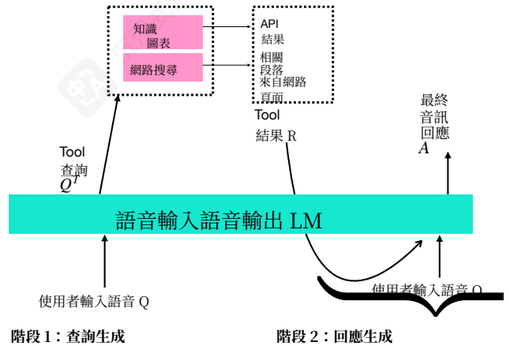
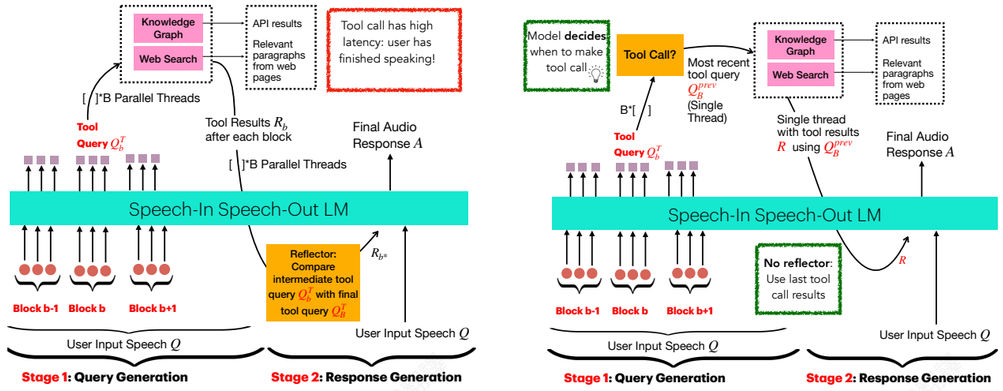


圖2 提出用兩階段推論方法整合工具使用的建議公式，應用於 E2E 語音輸入、語音輸出對話系統。



(a) Fixed-Interval Streaming RAG. **(b)** Model-Triggered Streaming RAG.

Figure 3 Proposed formulations for streaming tool query generation, referred to as *Streaming RAG*, to minimize user-perceived latency in speech-in, speech-out systems. (a) Fixed-Interval Streaming RAG: tool calls are triggered at fixed intervals and evaluated by a reflector module. (b) Model-Triggered Streaming RAG: the model autonomously decides when to make tool calls, eliminating the need for a reflector and directly utilizing the most recent tool results for response generation.

interacting with text-based APIs, but also preserves the key advantages of speech-in speech-out systems, mitigating error propagation and enabling the model to capture non-phonemic information (such as prosody and speaker intent) more effectively.

3.2 Streaming RAG

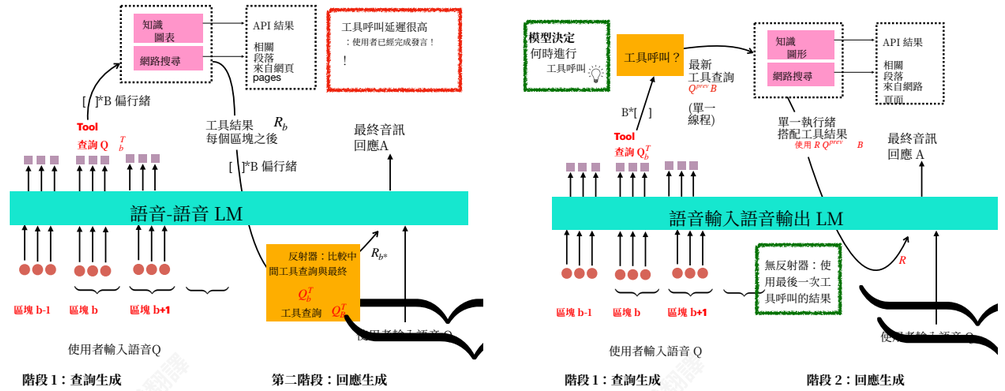
RAG-based systems, as proposed in S. 3.1, can significantly improve factual accuracy by incorporating external tools. However, these tool calls often introduce substantial latency, which is particularly problematic in speech-in, speech-out applications where users expect rapid, conversational responses and even brief silences can disrupt the natural flow of dialogue. One way to address this challenge lies in the nature of audio inputs, which arrive as a continuous stream. This streaming property enables tool calls to be initiated before the user has finished speaking, offering a unique opportunity to mitigate latency.

To minimize user-perceived latency, we introduce *Streaming RAG*: the first framework to generate and issue tool queries in parallel as audio input is received. This novel approach is built on three key design components: ① Trigger: When to initiate a new tool query; ② Threads: The number of parallel tool query threads; ③ Reflector: The module that determines whether intermediate tool results are sufficient for generating the final output. By exploring different design choices for each component, we introduce two complementary approaches for streaming tool query generation in the following subsections: a fixed-interval trigger method and a model-based trigger method.

3.2.1 Fixed-Interval Streaming RAG

In this approach, the *trigger* is set to fire tool calls at fixed chunk intervals during audio input. The input speech Q is divided into a sequence of B blocks, $Q = \{Q_b \mid b = 1, \ldots, B\}$, with each block containing N_{block} frames. To approximate $P(Q^T|Q)$ as described in S. 3.1, we follow a block-wise prediction strategy. In this approach, after processing each audio block b, the model predicts a tool query $\hat{Q}_b^T$ by conditioning on the input speech accumulated up to block b, specifically $Q_{1:b}$:

$$\hat{Q}_b^T = \arg\max_{Q_b^T} P(Q_b^T|Q_{1:b}) \tag{1}$$



(a) 固定間隔串流 RAG。 (b) 模型觸發串流 RAG。

圖 3 建議的串流工具查詢生成公式，稱為串流 RAG，以最小化語音輸入、語音輸出系統中的使用者感知延遲。(a) 固定間隔串流 RAG：工具呼叫在固定間隔觸發，並由反射模組評估。(b) 模型觸發串流 RAG：模型自主決定何時進行工具呼叫，無需反射模組，並直接利用最新的工具結果進行回應生成。

與基於文字的API互動，也保留了語音輸入-語音輸出系統的關鍵優勢，減輕錯誤傳播，並使模型能更有效地捕捉非音素信息（例如語調與演講者意圖）。

3.2 串流 RAG

基於 RAG 的系統，如 3.1 節所述，可透過整合外部工具顯著提高事實準確性。然而，這些工具呼叫通常會引入顯著的延遲，這在語音輸入、語音輸出應用中尤為問題，因為使用者期望快速、對話式的回應，甚至短暫的沉默都可能打斷對話的自然流動。解決這個挑戰的一種方法在於音頻輸入的本質，它以持續串流的形態到來。這種串流特性允許在使用者尚未結束發言前觸發工具呼叫，提供了一個獨特的機會來減緩延遲。

為了最小化使用者感知的延遲，我們引進了 Streaming RAG：作為第一個在接收音頻輸入時能夠並行生成並發出工具查詢的框架。這種創新的方法基於三個關鍵設計組件：1 觸發器：何時啟動新的工具查詢；2 線程：並行工具查詢線程的數量；3 反射器：判斷中間工具結果是否足夠用於生成最終輸出的模組。透過探索每個組件的不同設計選項，我們在以下小節中引進了兩種互補的方法，用於流式傳輸工具查詢生成：固定間隔觸發器方法和基於模型的觸發器方法。

3.2.1 固定間隔 Streaming RAG

在本方法中，觸發器設定為在音訊輸入期間以固定塊間隔觸發工具呼叫。輸入的語音 Q 被分成一系列的 B 塊， $Q = \{Q_b \mid b = 1, \ldots, B\}$，每個塊包含 N_{block}個框架。為了近似S. 3.1中描述的 $P(Q^T|Q)$ ，我們採用塊狀預測策略。在此方法中，在處理每個音訊塊 b後，模型透過條件化輸入到塊b積累的語音，具體是 $Q_{1:b}$，來預測工具查詢ˆ Q_b^T：

$$\hat{Q}_b^T = \arg\max_{Q_b^T} P(Q_b^T|Q_{1:b}) \tag{1}$$

This strategy results in B parallel tool call threads running simultaneously (see Figure 3a), where each thread generates a tool query prediction $\hat{Q}_b^T$ for its corresponding block $b \in [1, B]$. The tool queries $\hat{Q}_b^T$ generated after each block are then stored in cache. Given the high latency of tool calls (See T. 3), users typically complete their utterances before tool responses are ready. Upon utterance completion, an explicit *reflector* module "reflect()" evaluates the cached intermediate queries $\hat{Q}_b^T$ against final query $\hat{Q}_B^T$ to determine whether an early intermediate tool call provides sufficient information to answer the user's question Q. The reflector module systematically evaluates all intermediate queries in the cache and identifies the *earliest* sufficient tool call $b^\star$ where the intermediate query $\hat{Q}_b^T$ will give the same result as final query $\hat{Q}_B^T$ as shown:

$$b^\star = \min\{b \in [1, B] \text{ where reflect}(\hat{Q}_b^T, \hat{Q}_B^T) = \text{True}\}. \tag{2}$$

All subsequent parallel tool calls after $b^\star$ are promptly terminated, and the retrieved results $R_{b^\star}$ from this intermediate call $\hat{Q}_{b^\star}^T$ are used to generate the final spoken response A by maximizing the posterior distribution $P(A|Q, R_{b^\star})$ (instead of $P(A|Q, R)$ in S. 3.1). We employ a reflector module that uses simple yet effective heuristics: (a) if the top 5 web documents for an intermediate web query match those of the final web query, and (b) if the KG results for intermediate and final KG queries are identical. These heuristics ensure that the information retrieved from an early tool call using $\hat{Q}_{b^\star}^T$ is consistent with what would have been obtained by waiting for the final tool call using $\hat{Q}_B^T$, thereby providing a strong quality guarantee. Since most tool call latency arises from chunking and reranking the chunks of web documents, these checks enable significant latency savings without any compromise in model performance. A key advantage of this strategy is its plug-and-play nature: it requires no additional post-training for speech-in, speech-out models and can be directly applied at inference time across a variety of architectures. However, there are important considerations. First, generating parallel tool calls at every fixed interval (Eq. 1) increases computational overhead, which may pose challenges for deployment on resource-constrained devices such as wearables devices. Second, reliance on an external reflector module (Eq. 2) to determine the sufficiency of intermediate tool calls may limit the extent of achievable latency improvements.

3.2.2 Model-Triggered Streaming RAG

To address the limitations of Fixed-Interval Streaming RAG and further optimize both efficiency and responsiveness, we propose a more adaptive approach: *Model-Triggered Streaming RAG*. Here, the *trigger* is learned: the model is trained to autonomously determine the optimal moments to initiate tool queries, issuing a query only when it encounters new or additional information as illustrated in Figure 3b. In this formulation, the model receives user input in fixed chunk intervals as before and intelligently determines whether a tool call is needed after each block b. The model can either: ① Predict NO_QUERY if a new tool query is unnecessary, or ② Generate a new tool query. To make this decision, the model conditions on both the accumulated input speech $Q_{1:b}$ (see Eq. 1) as well as most recent tool query $\hat{Q}_b^{\text{prev}} = \hat{Q}^T_{\max\{\, i<b:\hat{Q}_i^T \neq \text{NO_QUERY}\,\}}$ as shown:

$$\hat{Q}_b^T = \arg\max_{Q_b^T} P(Q_b^T | Q_{1:b}, \hat{Q}_b^{\text{prev}}) \tag{3}$$

When a new query $\hat{Q}_b^T \neq$ NO_QUERY, the system immediately terminates any ongoing tool calls for the previous query $\hat{Q}_b^{\text{prev}}$, ensuring that only a *single tool call thread* runs at any given time. (Examples of generated $\hat{Q}_b^T$ are provided in T. 8 in the Appendix.) This approach offers several key advantages. First, it effectively eliminating redundant parallel threads and significantly reducing computational overhead. This is especially important for deployment on resource-constrained devices. Second, this formulation removes the need for an external *reflector* (Eq. 2) module. The model confidently relies on the results R from the most recent tool call using $\hat{Q}_B^{\text{prev}}$ to generate the spoken response A by maximizing $P(A|Q, R)$, reducing system complexity.

Post-training: To train the model, we transform text-based tool usage benchmarks into spoken format (See S. G). Word-level timestamps are computed using a pre-trained ASR model. For each partial ASR transcript X_b^{asr} up to block b, we generate corresponding queries for each tool $\overline{Q_b^T}$ using an LLM as pseudo ground truth (GT). To create effective training labels that teach the model when to trigger new queries, we employ a similarity-based labeling strategy where we compare the current pseudo GT query $\overline{Q_b^T}$ with the most recent non-empty tool query label $\hat{Q}_b^{\text{prev}}$ (see Eq. 3) before block b. Our labeling function assigns the training label

此策略導致 B 平行工具呼叫執行緒同時運行（見圖3a），其中每個執行緒為其對應的塊 $b \in [1, B]$生成工具查詢預測ˆ Q_b^T。每個塊生成後的工具查詢 $\hat{Q}_b^T$則存入快取。由於工具呼叫的高延遲（見T. 3），用戶通常在工具響應準備好之前就完成了他們的發言。在發言完成後，一個明確的反射模組 "reflect()" 會評估快取的中間查詢 $\hat{Q}_b^T$與最終查詢 $\hat{Q}_B^T$ ，以判斷早期中間工具呼叫是否提供足夠的資訊來回答用戶的問題 Q。反射模組系統地評估快取中的所有中間查詢，並識別最早足夠的工具呼叫 $b^\star$ ，其中中間查詢 $\hat{Q}_b^T$將與最終查詢 $\hat{Q}_B^T$ 產生相同結果，如圖所示：

$$b^\star = \min\{b \in [1, B] \text{ where reflect}(\hat{Q}_b^T, \hat{Q}_B^T) = \text{True}\}. \tag{2}$$

在 $b^\star$ 之後的所有平行工具呼叫都會立即終止，並且從這個中間呼叫中取得的結果 $R_{b^\star}$ 會用來生成最終的語音回應 A ，方法是最大化後驗分佈$P(A|Q, R_{b^\star})$（而不是 $P(A|Q, R)$ 在 S. 3.1 中）。我們採用一個反射模組，它使用簡單但有效的啟發式方法：(a) 如果中間網頁查詢的前 5 個網頁文件與最終的網頁查詢匹配，以及 (b) 如果知識圖形（KG）結果對於中間和最終的 KG 查詢是相同的。這些啟發式方法確保從早期工具呼叫使用 $\hat{Q}_{b^\star}^T$ 所取得的資訊與等待使用 $\hat{Q}_B^T$ 的最終工具呼叫所取得的資訊一致，從而提供強大的品質保證。由於大多數工具呼叫延遲都來自於網頁文件塊的分割和重新排序，這些檢查能夠在不犧牲模型性能的情況下顯著節省延遲。這種策略的一個主要優勢是其即插即用的特性：它不需要對語音輸入、語音輸出模型進行額外的後訓練，並且可以在各種架構的推理時間直接應用。然而，有重要的考慮因素。首先，在每個固定間隔生成平行工具呼叫（公式 1）會增加計算開銷，這可能會對在資源受限的裝置（例如可穿戴裝置）上部署帶來挑戰。其次，依賴外部反射模組（公式 2）來判斷中間工具呼叫的足夠性可能會限制可達到的延遲改善程度。

3.2.2 模型觸發式串流 RAG

為了彌補固定間隔串流 RAG 的限制，並進一步優化效率與反應速度，我們提出更具適應性的方法：模型觸發式串流 RAG。在此方案中，觸發機制是透過學習獲得：模型被訓練成能自主判斷啟動工具查詢的最佳時刻，僅在遇到新或額外資訊時才發出查詢，如圖 3b 所示。在此架構中，模型依舊以固定塊距接收用戶輸入，並在每個塊 b 後智能判斷是否需要呼叫工具。模型可以選擇：1 若無需新的工具查詢，則預測 NO_QUERY；或 2 生成新的工具查詢。為了做出此決策，模型同時基於累積的輸入語音 $Q_{1:b}$(參見公式 1) 以及最新的工具查詢 $\hat{Q}_b^{\text{prev}} = \hat{Q}^T_{\max\{\, i<b:\hat{Q}_i^T \neq \text{NO_QUERY}\,\}}$ 進行條件判斷：

$$\hat{Q}_b^T = \arg\max_{Q_b^T} P(Q_b^T | Q_{1:b}, \hat{Q}_b^{\text{prev}}) \tag{3}$$

當有新的查詢 $\hat{Q}_b^T \neq$ NO_QUERY 時，系統會立即終止先前查詢 $\hat{Q}_b^{\text{prev}}$ 的任何進行中的工具呼叫，確保在任何時候都只有單一工具呼叫執行緒在運作。(附件中的 T. 8 提供了產生 $\hat{Q}_b^T$ 的範例。) 這種方法提供了幾個關鍵優勢。首先，它能有效消除冗餘的平行執行緒，顯著降低計算開銷。這對於部署在資源受限的設備上尤其重要。其次，這種公式移除了對外部反射器 (Eq. 2) 模塊的需求。模型自信地依賴最新工具呼叫的結果 R，使用 $\hat{Q}_B^{\text{prev}}$ 來生成口語回應 A，透過最大化 $P(A|Q, R)$，降低系統複雜度。

訓練後：為了訓練模型，我們將基於文本的工具使用基準轉換為語音格式（見 S. G）。詞級時間戳使用預訓練的 ASR 模型計算。對於每個到達區塊 b 的部分 ASR 語音稿 X_b^{asr}，我們使用 LLM 作為虛擬實際標準（GT）為每個工具 Q_b^T 生成對應的查詢。為了創建有效的訓練標籤，教導模型何時觸發新的查詢，我們採用基於相似性的標籤策略，在區塊 b 之前，我們將當前虛擬 GT 查詢 Q_b^T 與最近非空的工具查詢標籤ˆ Q_b^{prev} （見公式 3）進行比較。我們的標籤函數分配訓練標籤

Table 1 Performance comparison of accuracy, first-token latency, and latency savings (as a percentage of tool use latency, which is 3.37 seconds as reported in Table 3) for three models—Qwen2.5-7B, OpusLM, and Kimi Audio—across three settings: Closed Book (without tool usage), Open Book (with tool usage), and Streaming RAG (*Model-Triggered Streaming RAG*). We evaluate all models on both the AudioCRAG-Synthetic (Syn.) and AudioCRAG-Human (Hum.). Streaming RAG is not applied to Kimi-Audio as it can handle only a restricted length of tool result references (S. F). *: OpusLM currently does not support taking tool result references in speech-out settings in zero-shot.

Setting	Ref length	Model	Accuracy		Latency			
					First Token (s)		% Savings	
			Syn.	Hum.	Syn.	Hum.	Syn.	Hum.
Closed Book	0	Qwen2.5-7B	11.1	13.1	1.34	1.24	✗	✗
	0	OpusLM	18.4	15.5	5.67	7.07	✗	✗
	0	Kimi Audio	16.7	16.0	0.85	0.89	✗	✗
Open Book (S. 3.1)	23K	Qwen2.5-7B	26.3	26.9	5.90	5.40	✗	✗
	15K	OpusLM*	0.0	0.0	9.05	10.44	✗	✗
	500	Kimi Audio	21.8	19.6	4.22	4.22	✗	✗
Streaming RAG (S. 3.2.2)	23K	Qwen2.5-7B	34.2	37.4	5.32	3.60	20.7%	53.4%
	15K	OpusLM	23.6	22.8	8.63	9.04	14.8%	41.5%

$\hat{Q}_b^T$ (Eq. 3) for the tool query after block b as follows: ① when the current query is sufficiently similar to the previous query (as determined by manually defined heuristics $f(\cdots)$), we assign the special label NO_QUERY to teach the model that no new tool call is needed. ② When the queries are sufficiently different, we assign the actual pseudo ground truth query $\overline{Q_b^T}$ as the label to teach the model to trigger a new tool call:

$$\hat{Q}_b^T = \begin{cases} \text{NO_QUERY}, & \text{if } f(\overline{Q_b^T}, \hat{Q}_b^{\text{prev}}) = \text{True}, \\ \overline{Q_b^T}, & \text{else .} \end{cases} \tag{4}$$

For KG queries, we assign a NO_QUERY label when the current query exactly matches the previous one. For web queries, we assign a NO_QUERY label if the top five retrieved documents for the current query remain unchanged from the previous query.

We employ a multi-task fine-tuning strategy targeting two key capabilities. First, we train the model on Streaming Tool Query Generation by optimizing $P(Q_b^T|Q_{1:b}, \hat{Q}_b^{\text{prev}})$ for $b \in [1, B]$, enabling intelligent decisions about when to trigger tool queries. Second, we fine-tune on Response Generation by optimizing $P(A|Q, R)$ (S. 3.1) to improve the intelligibility of the speech output.

An important aspect of our post-training is enhancing the model's ability to recover from errors in intermediate query predictions. For example, when presented with the audio question, "Who founded Rare Beauty in 2019?", we observed that an initial misinterpretation of $\hat{Q}_b^{\text{prev}}$, such as "Red Bull founder", can lead the model to subsequently generate NO_QUERY labels, effectively halting further attempts to retrieve the correct information. This issue arises because, during training, the model is always provided with correct previous query $\hat{Q}_b^{\text{prev}}$, whereas during inference, it may make errors due to partial utterances $Q_{1:b}$ being ambiguous. Thus the model lacks the ability to recover from such mistakes during inference. To overcome this, we introduce a novel strategy in which we deliberately inject negative samples during post-training by substituting the previous query $\hat{Q}_b^{\text{prev}}$ in Eq. 4 with incorrect ones Q_b^{neg}. Crucially, when we perform negative sampling, we fall back to the pseudo ground truth query $\overline{Q_b^T}$ as the training label:

$$(\hat{Q}_b^T, \hat{Q}_b^{\text{prev}}) = \begin{cases} (\hat{Q}_b^T, \hat{Q}_b^{\text{prev}}), & \text{with probability 0.9,} \\ (\overline{Q_b^T}, Q_b^{\text{neg}}), & \text{with probability 0.1.} \end{cases} \tag{5}$$

This approach explicitly teaches the model to recover from errors in intermediate query prediction, thereby maintaining accuracy (see T. 5 for ablation) while achieving latency savings.

表1 三個模型（Qwen2.5-7B、OpusLM 與 Kimi Audio）在閉卷（無工具使用）、開卷（有工具使用）及串流 RAG（模型觸發串流 RAG）三種設定下，精準度、首token延遲和延遲節省（以工具使用延遲的百分比表示，如表3所報告的3.37秒為基準）的性能比較。我們在 AudioCRAG-Synthetic (Syn.) 和 AudioCRAG-Human (Hum.) 上評估所有模型。串流 RAG 不適用於 Kimi-Audio，因其僅能處理有限長度的工具結果參考（S. F）。*：OpusLM 目前在零樣本設定中，於語音輸出設定下不支援使用工具結果參考。

設定	參考長度	模型	精確度		延遲			
					第一個 Token (s)		% 儲省	
			Syn.	Hum.	Syn.	Hum.	Syn.	Hum.
閉本書	0	Qwen2.5-7B	11.1	13.1	1.34	1.24	✗	✗
	0	OpusLM	18.4	15.5	5.67	7.07	✗	✗
	0	Kimi 音訊	16.7	16.0	0.85	0.89	✗	✗
開啟書籍 (S. 3.1)	23K	Qwen2.5-7B	26.3	26.9	5.90	5.40	✗	✗
	15K	OpusLM*	0.0	0.0	9.05	10.44	✗	✗
	500	Kimi 音訊	21.8	19.6	4.22	4.22	✗	✗
串流 RAG (S. 3.2.2)	23K	Qwen2.5-7B	34.2	37.4	5.32	3.60	20.7%	53.4%
	15K	OpusLM	23.6	22.8	8.63	9.04	14.8%	41.5%

$\hat{Q}_b^T$（公式 3）給區塊 b 後的工具查詢，如下所示：1 當當前查詢與先前查詢足夠相似時（由手動定義的啟發式方法 $f(\cdots)$ 確定），我們分配特殊標籤 NO_QUERY 教導模型不需要新的工具調用。2 當查詢足夠不同時，我們分配實際的虛擬實際標準查詢 Q_b^T 作為標籤，教導模型觸發新的工具調用：

$$\hat{Q}_b^T = \begin{cases} \text{NO_QUERY}, & \text{if } f(\overline{Q_b^T}, \hat{Q}_b^{\text{prev}}) = \text{True}, \\ \overline{Q_b^T}, & \text{else .} \end{cases} \tag{4}$$

針對KG查詢，當當前查詢與先前查詢完全一致時，我們會賦予NO_QUERY標籤。針對網路查詢，當當前查詢的前五個回傳文檔與先前查詢保持不變時，我們會賦予NO_QUERY標籤。

我們採用一種針對兩項關鍵能力的多任務微調策略。首先，我們透過優化 $P(Q_b^T|Q_{1:b}, \hat{Q}_b^{\text{prev}})$ 以針對 $b \in [1, B]$ 來在Streaming Tool Query Generation上訓練模型，從而能夠智能地決定何時觸發工具查詢。其次，我們透過優化 $P(A|Q, R)$ (S. 3.1) 在Response Generation上進行微調，以提升語音輸出的可理解性。

我們後訓練的一個重要方面是增強模型從中間查詢預測中的錯誤中恢復的能力。例如，當呈現音訊問題「Who founded Rare Beauty in 2019?」時，我們觀察到對 $\hat{Q}_b^{\text{prev}}$ 的初始錯誤解讀，例如「Red Bull founder」，可能導致模型隨後生成NO_QUERY標籤，有效地停止了進一步嘗試獲取正確信息的嘗試。這個問題的發生是因為，在訓練期間，模型總是獲得正確的先前查詢 $\hat{Q}_b^{\text{prev}}$ ，而在推論期間，由於部分發音 $Q_{1:b}$ 模糊不清，它可能會犯錯誤。因此，模型缺乏在推論期間從這種錯誤中恢復的能力。為了克服這個問題，我們在後訓練期間引入了一種新策略，通過將等式4中的先前查詢 $\hat{Q}_b^{\text{prev}}$ 替換為不正確的樣本 Q_b^{neg} 來故意注入負樣本。關鍵在於，當我們執行負樣本採樣時，我們會回退到虛擬真實查詢Q_b^T作為訓練標籤：

$$(\hat{Q}_b^T, \hat{Q}_b^{\text{prev}}) = \begin{cases} (\hat{Q}_b^T, \hat{Q}_b^{\text{prev}}), & \text{with probability 0.9,} \\ (\overline{Q_b^T}, Q_b^{\text{neg}}), & \text{with probability 0.1.} \end{cases} \tag{5}$$

此方法明確教導模型從中間查詢預測中的錯誤中恢復過來，從而維持準確性（見 T. 5 的消融實驗）同時實現延遲降低。

4 Experiment Setup

4.1 Evaluation Benchmarks

To rigorously evaluate our proposed approach, we construct comprehensive benchmark datasets featuring spoken queries paired with simulated tool interactions. We begin with the CRAG dataset (Yang et al., 2024), which form the basis for our spoken version of CRAG, which we term *AudioCRAG*. It consists of 2 distinct variants: ① **Audio CRAG Synthetic**: To generate spoken queries, we use our in-house TTS system. We then apply a rigorous filtering procedure which results in a high-quality set of 1,862 spoken queries, which we refer to as the *AudioCRAG-Synthetic* benchmark. ② **Audio CRAG Human**: To further enhance the realism and diversity of our evaluation, we introduce the AudioCRAG-Human benchmark, which consists of 618 human-recorded spoken queries. We will release Audio CRAG Human benchmark upon acceptance to supports the development of more natural and reliable voice assistants. Further details on the construction of these benchmarks are provided in Sec. C. We follow the CRAG setup to incorporate both web and KG-based tools, and adopt its robust evaluation methodology, as described in Secs. D and E. Additionally, we leverage a random subset of 16,000 questions from the text-based factual question answering dataset TriviaQA (Joshi et al., 2017) to post-train our speech-in, speech-out models, as detailed in Sec. G.

4.2 Evaluated SOTA Speech-in Speech-out Models

In this work, we present a comprehensive benchmark of three SOTA speech-in, speech-out conversational systems: Qwen-OMNI (Xu et al., 2025), Kimi-Audio (Ding et al., 2025) and OpusLM (Tian et al., 2025). We evaluate them under both tool-augmented and non-tool-augmented conditions. Further details about our experimental setup are provided in S. F.

We perform an ablation study (referred to as "Tool Integration" in T. 4) where we post-train the model on sequential query generation (i.e. $P(Q^T|Q)$ in S. 3.1) and output generation, to assess the impact of streaming RAG post-training versus standard post-training on final response generation. Additionally, we conduct an ablation study on open book setting (S. 3.1) using a self-cascade approach with a three-stage inference pipeline: (1) the audio question is used to generate a tool query Q^T (corresponding to the "Query Generation" stage described in S. 3.1); (2) the audio question, and tool results are combined to produce the final text output X^{res} by maximizing $P(X^{\text{res}}|Q, R)$; and (3) the audio question, and final text output are used to generate the final speech output A by optimizing $P(A|Q, X^{\text{res}})$. Since we teacher-force the text output to obtain the final speech output in stage (3), this self-cascade approach can only be applied to a "thinker-talker" architecture (eq. Qwen-OMNI) or Chain-of-Thought (CoT) style architectures (eg. OpusLM). The motivation for this ablation is to investigate whether the inclusion of RAG references R during the "Response Generation" stage (S. 3.1) affects the quality of the generated speech A.

5 Results

5.1 Impact of Tool Integration and Streaming RAG on SOTA Models

Table 1 provides a comprehensive performance comparison of three models, Qwen2.5-7B, OpusLM, and Kimi Audio, evaluated across three settings: Closed Book, Open Book, and Streaming RAG. All models are assessed on both the AudioCRAG-Synthetic (Syn.) and AudioCRAG-Human (Hum.). In the Closed Book setting, where models rely solely on their internal knowledge without access to external tools (reference length = 0), all models achieve accuracy scores below 20%. These results highlight the inherent limitations of closed-book approaches in handling complex queries. The Open Book setting, which provides models with access to external information, demonstrates the clear benefits of tool integration. Qwen2.5-7B and Kimi Audio's accuracy rises substantially, underscoring the value of leveraging external context. As expected, latency increases due to the additional overhead of retrieving information from external tools (See T. 3 for detailed latency analysis).

Most notably, our post-training approach to build *Model-Triggered Streaming RAG*, as described in S. 3.2.2, delivers significant advancements in both accuracy and efficiency. Qwen2.5-7B and OpusLM achieve significant

4 實驗設定

4.1 評估基準

為嚴格評估我們提出的方針，我們構建了包含口述查詢與模擬工具互動的全面基準數據集。我們從 CRAG 數據集（Yang 等人，2024）開始，該數據集構成我們口述版 CRAG 的基礎，我們稱之為 AudioCRAG。它包含 2 個不同的變體：1 Audio CRAG Synthetic：為了生成口述查詢，我們使用我們的內部 TTS 系統。接著我們應用嚴格的過濾程序，產生了一個高質量的 1,862 個口述查詢集，我們稱之為 AudioCRAG-Synthetic 基準。2 Audio CRAG Human：為了進一步提高評估的真實性和多樣性，我們引入了 AudioCRAG-Human 基準，它包含 618 個人聲錄製的口述查詢。我們將在接收後發布 Audio CRAG Human 基準，以支持更自然可靠的語音助理的開發。關於這些基準構建的更多細節，請參見 Sec. C。我們遵循 CRAG 設置來結合基於網絡和知識庫的工具，並採用其強大的評估方法，如 Secs. D 和 E 所述。此外，我們利用來自基於文本的事實性問答數據集 TriviaQA（Joshi 等人，2017）的隨機子集 16,000 個問題，來進行我們的語音輸入、語音輸出模型的後訓練，詳情請參見 Sec. G。

4.2 評估的SOTA語音-語音對話模型

在本研究中，我們提出三個SOTA語音輸入、語音輸出對話系統的全面基準：Qwen-OMNI (Xu等人，2025年)、Kimi-Audio (Ding等人，2025年) 和 OpusLM (Tian等人，2025年)。我們在工具增強和非工具增強的情況下評估它們。我們實驗設定的更多細節提供在S. F中。

我們進行了一項剪枝研究（在T. 4中稱為“工具整合”），其中我們在順序查詢生成（即S. 3.1中的 $P(Q^T|Q)$）和輸出生成上進行後訓練，以評估流式RAG後訓練與標準後訓練對最終響應生成的影響。此外，我們使用自級聯方法，採用三階段推理管道進行開放書籍設置（S. 3.1）的剪枝研究：(1) 使用音頻問題生成工具查詢 Q^T（對應於S. 3.1中描述的“查詢生成”階段）；(2) 將音頻問題和工具結果結合，通過最大化 $P(X^{\text{res}}|Q, R)$ 產生最終文本輸出X^{res}；(3) 使用音頻問題和最終文本輸出通過優化 $P(A|Q, X^{\text{res}})$ 生成最終語音輸出 A。由於我們在階段(3)中強制使用文本輸出來獲得最終語音輸出，因此這種自級聯方法只能應用於“思考者-談話者”架構（例如Qwen-OMNI）或思維鏈（CoT）風格架構（例如OpusLM）。進行此剪枝的動機是調查在“響應生成”階段（S. 3.1）期間包含RAG參考 R 是否會影響生成的語音 A 的質量。

5 個結果

5.1 工具整合與串流 RAG 對 SOTA 模型的影響

表 1 提供三個模型 Qwen2.5-7B、OpusLM 與 Kimi Audio 的綜合性能比較，評估設定為三種：封閉書籍、開放書籍與串流 RAG。所有模型均在 AudioCRAG-Synthetic (Syn.) 與 AudioCRAG-Human (Hum.) 上進行評估。在封閉書籍設定中，模型僅依賴其內部知識而不存取外部工具（參考長度 = 0），所有模型的準確率均低於 20%。這些結果突顯了封閉書籍方法在處理複雜查詢上的內在限制。開放書籍設定提供模型存取外部資訊，展現了工具整合的明顯效益。Qwen2.5-7B 與 Kimi Audio 的準確率顯著提升，強調了利用外部語境的價值。預期中，由於存取外部工具的額外開銷，延遲增加（詳見表 3 的延遲分析）。

最值得注意的是，我們的訓練後方法來構建 Model-Triggered Streaming RAG，如 S. 3.2.2 中所述，在準確性和效率方面都取得了顯著進步。Qwen2.5-7B 和 OpusLM 在兩個基準測試中都實現了顯著的準確性提升。

Table 2 Results on AudioCRAG-Synthetic for Qwen2.5-7B, OpusLM, and Kimi Audio comparing text vs. speech output across Closed Book, Open Book, and *Model-Triggered Streaming RAG*.

Setting	Ref length	Output	Model	Acc.
Closed Book	0	Text	Qwen2.5-7B	15.0
	0	Speech	Qwen2.5-7B	11.1
	0	Text	OpusLM	20.1
	0	Speech	OpusLM	18.4
	0	Text	Kimi Audio	24.2
	0	Speech	Kimi Audio	16.7
Open Book (S. 3.1)	23K	Text	Qwen2.5-7B	39.6
	23K	Speech	Qwen2.5-7B	26.3
	23K	Speech (self-cascade)	Qwen2.5-7B	33.8
	15K	Text	OpusLM	26.3
	15K	Speech	OpusLM	0.0
	15K	Speech (self-cascade)	OpusLM	21.2
	5K	Text	Kimi Audio	45.8
	500	Speech	Kimi Audio	21.8
Streaming RAG (S. 3.2.2)	23K	Text	Qwen2.5-7B	39.8
	23K	Speech	Qwen2.5-7B	34.2
	15K	Text	OpusLM	29.7
	15K	Speech	OpusLM	23.6

Table 3 First Token Latency breakdown, showing median (P50) and 90th percentile (P90) timings, for the Qwen2.5-7B on AudioCRAG-Synthetic.

Model	Setting	P	Latency (sec)			
			Tool Use Latency		Response Gen	Total
			Query Gen	Tool Results Gen		
Qwen2.5-7B	Open Book (S. 3.1)	P50	0.59	2.78	2.52	5.90
		P90	0.85	4.90	3.25	9.00
	Streaming RAG (S. 3.2.2)	P50	0.59	2.20	2.52	5.32
		P90	0.85	4.37	3.25	8.47

accuracy improvements across both benchmarks. While the absolute accuracy scores may appear low, they are consistent with the evaluation results observed in the CRAG benchmark. Notably, Qwen2.5-7B with *Model-Triggered Streaming RAG* achieves accuracy comparable to the open book performance of similarly sized LLMs reported in the original CRAG paper (i.e., 34.2 for Qwen2.5-7B vs. 32.1 for LLAMA-3 8B Instruct in (Yang et al., 2024)). Importantly, although post-training is performed exclusively on the synthetic dataset, we observe consistent and even greater improvements on the human-spoken benchmark, demonstrating the robustness and strong generalization capabilities of our method. This setting also introduces substantial latency savings compared to the Open Book configuration, with Qwen2.5-7B and OpusLM achieving 20.7% and 14.8% reductions in first-token latency on the synthetic benchmark, and even greater savings on the human benchmark [1]. These results demonstrate that our streaming RAG approach not only advances the accuracy of speech-in speech-out systems, but also optimizes response efficiency by enabling earlier and more effective prediction of tool queries.

表2 在AudioCRAG-Synthetic上針對Qwen2.5-7B、OpusLM和Kimi Audio的結果，比較文本與語音輸出在閉卷、開卷和模型觸發串流RAG中的表現。

設定	參考長度	輸出	模型	Acc.
閉卷	0	Text	Qwen2.5-7B	15.0
	0	語音	Qwen2.5-7B	11.1
	0	Text	OpusLM	20.1
	0	語音	OpusLM	18.4
	0	Text	Kimi Audio	24.2
	0	語音	Kimi Audio	16.7
Open Book (S. 3.1)	23K	Text	Qwen2.5-7B	39.6
	23K	語音	Qwen2.5-7B	26.3
	23K	語音 (自我疊加)	Qwen2.5-7B	33.8
	15K	Text	OpusLM	26.3
	15K	語音	OpusLM	0.0
	15K	語音 (自我疊加)	OpusLM	21.2
	5K	Text	Kimi Audio	45.8
	500	語音	Kimi Audio	21.8
串流 RAG (S. 3.2.2)	23K	Text	Qwen2.5-7B	39.8
	23K	語音	Qwen2.5-7B	34.2
	15K	Text	OpusLM	29.7
	15K	演講	OpusLM	23.6

Table3 第一次令牌延遲分解，顯示中位數（P50）和90百分位數（P90）的時間，針對Qwen2.5-7B在AudioCRAG-Synthetic上。

模型	設定	P	延遲（秒）			
			工具使用延遲		回應生成	總計
			查詢生成	工具結果生成		
Qwen2.5-7B	開啟書籍 (S. 3.1)	P50	0.59	2.78	2.52	5.90
		P90	0.85	4.90	3.25	9.00
	串流 RAG (S. 3.2.2)	P50	0.59	2.20	2.52	5.32
		P90	0.85	4.37	3.25	8.47

雖然絕對準確性分數可能看起來較低，但它們與 CRAG 基準測試中觀察到的評估結果一致。值得注意的是，使用 Model-Triggered Streaming RAG 的 Qwen2.5-7B 实現了與原始 CRAG 文章中報告的類似大小 LLM 的開放書性能相當的準確性（即，Qwen2.5-7B 為 34.2，對於 LLAMA-3 8B Instruct 則為 32.1，參考 (Yang et al., 2024)）。重要的是，儘管訓練後優化完全在合成數據集上進行，我們在人工發音基準測試上觀察到一致甚至更大的進步，這證明了我們方法的穩健性和強泛化能力。這種設定與開放書配置相比，也引入了顯著的延遲節省，Qwen2.5-7B 和 OpusLM 在合成基準測試上第一個 token 的延遲減少了 20.7% 和 14.8%，在人工基準測試上的節省甚至更大 [1]。這些結果表明，我們的流式 RAG 方法不僅提高了語音輸入語音輸出的系統準確性，還通過實現更早、更有效的工具查詢預測來優化響應效率。

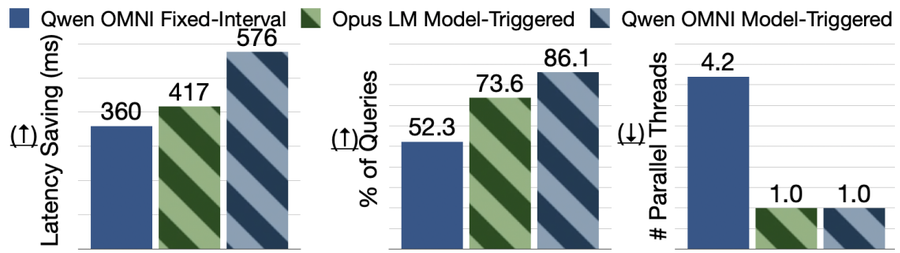


Figure 4 Average latency savings (Mean) by streaming RAG approaches (S. 3.2.)

5.2 Analysis: Modality Gap Between Speech and Text Output

Table 2 provides a comparative evaluation of the three SOTA models in generating either text or speech outputs from speech inputs, both with and without the integration of external tool results. In the absence of tool results (Ref length = 0), all models achieve higher accuracy when generating text outputs compared to speech outputs. The incorporation of tool results generally leads to improved text generation performance, with Kimi Audio achieving the highest accuracy. In contrast, the accuracy for speech output remains consistently lower across all models and conditions. The self-cascade approach, in which the model first generates an intermediate text response before producing the final speech output, provides moderate improvements in speech output accuracy for both Qwen2.5-7B and OpusLM. However, it still underperforms compared to the text-out baseline, primarily due to errors in accurately generating answers involving uncommon entity nouns. Overall, these findings underscore a persistent challenge in direct speech generation, particularly when tool results are integrated, as this appears to negatively impact the quality of generated speech responses. Our *Model-Triggered Streaming RAG* demonstrates clear advantages: it maintains comparable performance for text output and delivers substantial improvements for speech output, even outperforming the self-cascade approach. These results underscore the effectiveness of post-training in overcoming challenges in direct speech generation in tool-augmented scenarios.

5.3 Analysis of Latency Bottlenecks in Tool-Integrated Speech Dialogue

Table 3 provides a comprehensive breakdown of latency measurements for the Qwen2.5-7B speech-to-speech model, evaluated in both the Open Book setting and our proposed *Model-Triggered Streaming RAG* setting on AudioCRAG-Synthetic. We report both median (P50) and 90th percentile (P90) values for each stage of the processing pipeline. The latency is decomposed into three main components: tool query generation, tool result generation, and speech response synthesis, with measurements provided for both the first token outputs (We also provide latency measurements for last token output in T. 9). For both tool query and tool result generation, it is assumed that all tools are accessed in parallel; thus, the reported latency corresponds to the maximum query or result generation time among all tools. The majority of this latency arises from leveraging external web pages, which introduces significant delays, most notably, increasing the first token latency by 2.3x in Open Book Setting. Notably, our *Model-Triggered Streaming RAG* setting enables early generation of tool results, successfully reducing P50 first token latency by 9.8% and tool use latency by 20.7%.

5.4 Ablation Study

Streaming RAG approaches: Figure 4 highlights the substantial latency savings enabled by the *Model-Triggered Streaming RAG* (S. 3.2.2), compared to the *Fixed-Interval Streaming RAG* (S. 3.2.1). Three key metrics are evaluated: overall latency savings, the percentage of queries benefiting from reduced latency, and the number of parallel threads required. Even without any post-training, the *Fixed-Interval Streaming RAG* approach

[1]Our latency calculations on synthetic audio exclude end-point detection latency, which is required in all production systems. Since our streaming RAG approach enables processing without waiting for end-point detection, including this factor would further amplify the observed latency savings, as observed in higher latency savings for human-spoken audio, where endpoint detection errors often introduce trailing silence.

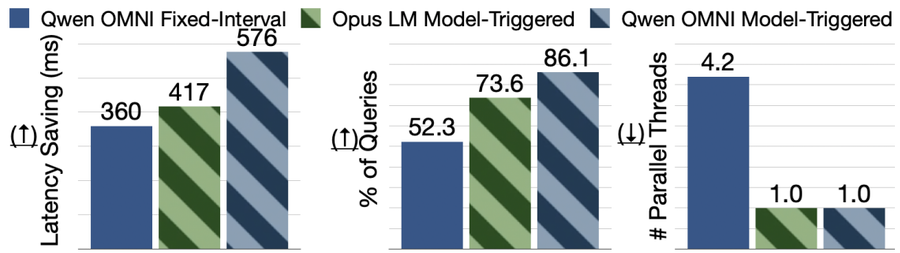


圖4 流式RAG方法平均延遲節省（平均）(S. 3.2.)

5.2 分析：語音與文本輸出的模態差距

表2 提供了三個SOTA模型在從語音輸入生成文本或語音輸出時的比較評估，包括與否整合外部工具結果。在沒有工具結果（參考長度 = 0）的情況下，所有模型在生成文本輸出時的準確率均高於生成語音輸出時。整合工具結果通常能提高文本生成性能，其中Kimi Audio達到最高準確率。相比之下，所有模型和條件下的語音輸出準確率始終保持較低。自適應級聯方法，即模型首先生成中間文本響應，然後產生最終語音輸出，對Qwen2.5-7B和OpusLM的語音輸出準確率提供了中度提升。然而，它仍低於文本輸出基線，主要是因為在準確生成涉及不常見實體名詞的答案時出錯。總體而言，這些發現突顯了直接語音生成的持續性挑戰，特別是在整合工具結果的情況下，因為這似乎會對生成的語音響應質量產生負面影響。我們的Model-Triggered Streaming RAG展現了明顯優勢：它在文本輸出方面維持相當性能，並為語音輸出帶來顯著提升，甚至超過了自適應級聯方法。這些結果突顯了訓練後優化的有效性，在工具增強場景中克服直接語音生成挑戰。

5.3 工具整合語音對話中的延遲瓶頸分析

表3提供了Qwen2.5-7B語音到語音模型在Open Book設定和我們提出的Model-Triggered Streaming RAG設定下，針對AudioCRAG-Synthetic的延遲測量綜合分解。我們報告了每個處理流程階段的中位數（P50）和90百分位數（P90）值。延遲被分解為三個主要組成部分：工具查詢生成、工具結果生成和語音響應合成，並為第一個詞元輸出（我們也在T. 9中提供了最後一個詞元輸出的延遲測量）。對於工具查詢和工具結果生成，假設所有工具都是並行訪問的；因此，報告的延遲對應於所有工具中的最大查詢或結果生成時間。這部分延遲的大多數來源於利用外部網頁，這引入了顯著的延遲，最為突出的是，在Open Book設定中將第一個詞元延遲增加了2.3倍。值得注意的是，我們的Model-Triggered Streaming RAG設定實現了工具結果的早期生成，成功將P50第一個詞元延遲降低了9.8%，並將工具使用延遲降低了20.7%。

5.4 損毀研究

串流 RAG 方法：圖 4 突顯了與固定間隔串流 RAG (S. 3.2.1) 相比，模型觸發串流 RAG (S. 3.2.2) 所能實現的顯著延遲節省。評估了三個關鍵指標：總體延遲節省、受益於延遲降低的查詢百分比，以及所需的平行線程數量。即使沒有任何後訓練，固定間隔串流 RAG 方法也

[1]我們在合成音訊上的延遲計算排除了端點偵測延遲，這是在所有生產系統中都需要的因素。由於我們的串流 RAG 方法能夠在等待端點偵測的情況下處理，包含這個因素將會進一步放大觀察到的延遲節省，正如人類發聲音訊中延遲節省更高所顯示的那樣，其中端點偵測錯誤常會引入尾隨靜音。

Table 4 Comparison of speech-to-speech model under different post-training conditions

Post-Training	Ref length	Post Train Data	Model	Acc.
Tool Integration (S. 4.2)	15K	16K	OpusLM	22.4
Streaming RAG (S. 3.2.2)	15K	16K	OpusLM	23.6
Tool Integration (S. 4.2)	23K	16K	Qwen2.5-7B	34.9
Streaming RAG (S. 3.2.2)	23K	16K	Qwen2.5-7B	34.2

Table 5 Ablation Results for Negative Sampling Strategy in *Model-Triggered Streaming RAG* (S. 3.2.2).

Scenario	Ref Length	Post Train Data	Output	Model	Acc.
Open Book	15K	0	Text	Qwen2.5-7B	39.6
Post-train (S. 3.2.2)	15K	16K	Text	Qwen2.5-7B	39.8
- Negative sampling	15K	16K	Text	Qwen2.5-7B	36.5

already reduces tool usage latency (3.37s in T. 3) by 10.7% for Open Book Qwen-OMNI, demonstrating its flexibility and plug-and-play compatibility with any existing speech-in speech-out model. The *Model-Triggered Streaming RAG* method, utilizing Qwen-OMNI, consistently delivers superior performance. It achieves greater average latency reductions and benefits a higher proportion of queries with improved response times. Notably, *Model-Triggered Streaming RAG* require only a single parallel thread, representing a significant advancement in resource efficiency compared to the *Fixed-Interval Streaming RAG* approach, which demands multiple parallel threads. These findings highlight the effectiveness of *Model-Triggered Streaming RAG* in not only minimizing latency, validating our intuition from Section 3.2.2, but also in optimizing system efficiency and resource utilization.

Post-training Strategies: Table 4 presents performance comparison under different post-training conditions. Incorporating streaming tool query generation during post-training (S. 3.2.2) results in comparable performance for both models. These results suggest that *Model-Triggered Streaming RAG* can be integrated into post-training process without negatively impacting model performance.

Ablation Results for Negative Sampling Strategy: Table 5 presents the ablation results evaluating the impact of deliberately injecting negative samples during post-training (Eq. 5). The findings highlight that, without negative sampling, streaming tool query generation can reduce final accuracy in text output settings, primarily due to errors in final query generation, as detailed in S. 3.2.2. In contrast, our negative sampling approach significantly enhances the model's robustness, enabling it to recover from intermediate prediction errors. This leads to consistently high accuracy while also achieving notable latency reductions (Tab. 3).

6 Conclusion and Discussion

In this work, we introduced the first comprehensive approach for integrating external tool usage directly into E2E speech-in, speech-out dialogue systems. By incorporating *Model-Triggered Streaming RAG* pipeline, we further enhanced the model's ability to leverage retrieved information and autonomously decide when to trigger new tool queries, resulting in improved accuracy and responsiveness. Empirical evaluation on the newly introduced AudioCRAG benchmark demonstrated that tool integration can more than double factual question answering accuracy compared to closed-book models. Additionally, our streaming RAG approach achieved a 20% reduction in tool usage latency, thereby preserving natural conversational flow. Overall, our contributions advance the state of the art in spoken dialogue systems by enabling accurate, real-time, and tool-augmented voice assistants. In addition, we are committed to open science and will release our training code and Audio CRAG Human benchmark to support future research and development in this area.

表4 不同後訓練條件下的語音轉語音模型比較

後訓練	參考長度	後訓練數據	模型	Acc.
工具整合 (S. 4.2)	15K	16K	OpusLM	22.4
串流 RAG (S. 3.2.2)	15K	16K	OpusLM	23.6
工具整合 (S. 4.2)	23K	16K	Qwen2.5-7B	34.9
串流 RAG (S. 3.2.2)	23K	16K	Qwen2.5-7B	34.2

表5 模型觸發串流RAG中負向採樣策略的消融結果 (S. 3.2.2)

場景	參考長度	訓練後數據	輸出	模型	Acc.
開啟書本	15K	0	Text	Qwen2.5-7B	39.6
後訓練 (S. 3.2.2)	15K	16K	Text	Qwen2.5-7B	39.8
- 負面採樣	15K	16K	Text	Qwen2.5-7B	36.5

已經將工具使用延遲降低 10.7%（在 T. 3 中為 3.37 秒），對 Open Book Qwen-OMNI 顯示其靈活性和即插即用與任何現有語音輸入語音輸出模型的兼容性。使用 Qwen-OMNI 的 Model-Triggered Streaming RAG 方法，始終提供卓越的性能。它實現了更高的平均延遲降低，並使更高比例的查詢受益於改善的響應時間。值得注意的是，Model-Triggered Streaming RAG 只需要單個平行線程，與需要多個平行線程的 Fixed-Interval Streaming RAG 方法相比，代表了資源效率的重大進步。這些發現突顯了 Model-Triggered Streaming RAG 在不僅最小化延遲（驗證了我們來自第 3.2.2 節的直覺），而且在優化系統效率和資源利用率方面的有效性。

後訓練策略：表4顯示了在不同後訓練條件下的性能比較。在後訓練期間加入流式工具查詢生成（S. 3.2.2）可讓兩個模型的性能相當。這些結果表明，模型觸發流式RAG可以整合到後訓練過程中，而無法對模型性能產生負面影響。

負採樣策略的消融實驗結果：表5顯示了評估在後訓練期間故意注入負採樣（公式5）影響的消融實驗結果。研究結果強調，若沒有負採樣，流式工具查詢生成會降低文本輸出設定中的最終準確率，主要原因是最終查詢生成中的錯誤，詳見S. 3.2.2。相比之下，我們的負採樣方法顯著提升了模型的穩定性，使其能從中間預測錯誤中恢復。這有助於保持持續的高準確率，同時也實現了顯著的延遲降低（表3）。

6 結論與討論

在本工作中，我們首次提出了一種將外部工具使用直接整合到端到端語音輸入、語音輸出對話系統的方法。透過整合模型觸發流式RAG流程，我們進一步提升了模型利用檢索信息並自主決定何時觸發新工具查詢的能力，從而提高了準確性和反應速度。在剛引進的AudioCRAG基準測試中，實證評估顯示工具整合可讓事實性問題回答準確率比封閉式模型高出兩倍以上。此外，我們的流式RAG方法將工具使用延遲降低了20%，從而保留了自然的對話流程。總體而言，我們的貢獻推進了語音對話系統的先進技術，實現了準確、實時且工具增強的語音助理。此外，我們致力於開放科學，將釋出我們的訓練程式碼和Audio CRAG Human基準，以支持此領域的未來研究與發展。

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Appendix

A Benchmarking Text Dialogue Systems for Tool Usage

Recent advances in benchmarking text-based dialogue systems for tool usage (Chen et al., 2023b; Ouyang et al., 2025; Cheng and Dou, 2025; Cohen et al., 2025; Xiong et al., 2024a; Vu et al., 2023; Xiong et al., 2024b; Peng et al., 2024; Su et al., 2025; Ni et al., 2025) have primarily focused on evaluating factual question answering and task completion within simulated environments. The CRAG benchmark (Yang et al., 2024) is a leading example, featuring 4,409 question-answer pairs and providing mock APIs for both web and knowledge graph (KG) search. CRAG supports a range of KG and web retrieval tasks, and highlights key challenges such as hallucinations in retrieval-augmented generation (RAG) and the importance of leveraging KGs and search ranking to improve factual accuracy. Evaluation is conducted automatically using two LLM judges. SimpleQA (Wei et al., 2024) is another widely adopted benchmark, designed to assess language models on short, fact-seeking questions. With 4,326 adversarially collected questions spanning diverse topics and a straightforward grading scheme based on single, indisputable answers, SimpleQA provides a robust testbed for factual accuracy. Moving beyond question answering, WebArena (Zhou et al., 2024) offers a simulated environment for evaluating dialogue agents on web-based tasks using fully functional websites, enabling assessment of more complex, action-oriented behaviors. While these benchmarks have significantly advanced the evaluation of tool-augmented dialogue systems, they remain largely limited to text-based interactions and do not fully address the unique challenges presented by speech-in speech-out systems.

B Multimodal Benchmarks for Tool Usage

Recent benchmarks Mei et al. (2025); Yu et al. (2025); Luo et al. (2024) have extended tool-augmented dialogue evaluation to multimodal and longer-context scenarios. The m&m's benchmark (Ma et al., 2024) evaluates LLMs on multi-step, multi-modal tasks using a diverse set of 33 tools, including public APIs and multimodal models such as off-the-shelf automatic speech recognition (ASR) models, highlighting the potential for developing agents that leverage audio-based tools. CRAG_MM (Meta CRAG-MM Challenge Organizers, 2025) builds on the original CRAG benchmark by introducing visual question answering (QA) tasks that combine images and text-based queries, utilizing mock APIs for both image descriptions and web search. For video understanding and long-context reasoning, the Video Web Arena (Jang et al., 2025) benchmark evaluates multimodal agents on tasks involving 2,021 manually crafted tutorial videos. While these benchmarks advance the field by incorporating multimodal tools, they still do not evaluate systems in speech-in speech-out scenarios.

C Audio CRAG Benchmark

We begin with the CRAG dataset (Yang et al., 2024), which contains 2,706 text queries. Since these queries are not directly suitable for speech-based evaluation, we first identify those requiring adaptation before TTS conversion. Through careful manual inspection, we determine that queries containing dates or brackets benefit from rewriting to ensure naturalness and clarity in spoken form. In total, we identify 569 such queries and rewrite them using a large language model (LLAMA-4 Maverick). The resulting 569 rewritten queries, combined with the remaining original queries, form the basis for our spoken version of CRAG, which we term *AudioCRAG*. We follow the CRAG setup to incorporate web and KG-based tools and adopt its robust evaluation setup, as detailed in S. D and E.

Audio CRAG Synthetic: To generate spoken queries, we process all 2,706 queries through our in-house TTS system. We then apply a rigorous filtering procedure to remove queries with intelligibility issues, specifically excluding any utterances for which Whisper Radford et al. (2023) hypotheses exhibit a non-zero word error rate. We also remove utterances with suboptimal audio quality, as determined by UTMOS (Saeki et al., 2022) scores below 3.5. This results in a high-quality set of 1,862 spoken queries, which we refer to as the *AudioCRAG-Synthetic* benchmark.

附錄

工具使用的基準測試文對話系統

近期針對工具使用基準測試文對話系統的進展 (Chen et al., 2023b; Ouyang et al., 2025; Cheng and Dou, 2025; Cohen et al., 2025; Xiong et al., 2024a; Vu et al., 2023; Xiong et al., 2024b; Peng et al., 2024; Su et al., 2025; Ni et al., 2025) 主要集中在評估模擬環境中的事實性問答和任務完成。CRAG基準測試 (Yang et al., 2024) 是一個領先的範例，包含4,409個問答對，並提供模擬API供網路和知識圖形 (KG) 搜尋使用。CRAG支援多種KG和網路檢索任務，並強調關鍵挑戰，例如檢索增強生成 (RAG) 中的虛構現象，以及利用KG和搜尋排名來提高事實準確性的重要性。評估是透過兩個LLM裁判自動進行。SimpleQA (Wei et al., 2024) 是另一個廣泛採用的基準測試，設計用於評估語言模型在短、尋求事實的問題上。透過4,326個對抗收集的、涵蓋多樣主題的問題，以及基於單一、無爭議答案的直觀評分方案，SimpleQA提供了一個堅固的事實準確性測試平台。超越問答，WebArena (Zhou et al., 2024) 提供一個模擬環境，用於評估對話代理在網路基礎任務上的表現，使用完全功能的網站，使評估更複雜、以動作為導向的行為成為可能。雖然這些基準測試顯著推進了工具增強對話系統的評估，但它們仍然主要限於文字基礎互動，並未完全應對語音輸入語音輸出系統所提出的獨特挑戰。

工具使用的多模態基準場

近年來，Mei 等人 (2025)；Yu 等人 (2025)；Luo 等人 (2024) 已將工具增強對話評估擴展至多模態與長文本情境。m&m 的基準場 (Ma 等人，2024) 使用包含公開 API 與多模態模型（例如現成自動語音辨識 ASR 模型）的 33 個工具組，對 LLM 在多步驟、多模態任務上的表現進行評估，突顯了開發利用聲音工具的代理的潛力。CRAG_MM (Meta CRAG-MM 挑戰組織者，2025) 在原 CRAG 基準場上進行擴充，引入結合圖像與文字查詢的視覺問答 (QA) 任務，並使用模擬 API 進行圖像描述與網路搜尋。針對影片理解與長文本推理，Video Web Arena (Jang 等人，2025) 基準場對涉及 2,021 個人工設計的教學影片的多模態代理進行評估。雖然這些基準場透過整合多模態工具推進了該領域，但它們仍然沒有評估在語音輸入、語音輸出的情境下系統的表現。

C 音頻 CRAG 基準測試

我們從 CRAG 資料集 (Yang 等人，2024) 開始，其中包含 2,706 個文字查詢。由於這些查詢不直接適用於基於語音的評估，我們在 TTS 轉換之前首先識別需要調整的查詢。透過仔細的手動檢查，我們確定包含日期或括號的查詢受益於重寫，以確保在語音形式中的自然性和清晰度。總共，我們識別出 569 個這樣的查詢，並使用大型語言模型 (LLAMA-4 Maverick) 將其重寫。這些 569 個重寫的查詢，加上剩下的原始查詢，構成了我們語音版本的 CRAG，我們稱之為 AudioCRAG。我們遵循 CRAG 設置來整合網路和知識圖形工具，並採用其強大的評估設置，詳情請參見 S. D 和 E。

Audio CRAG Synthetic：為了生成語音查詢，我們將所有 2,706 個查詢通過內部 TTS 系統處理。接著，我們應用嚴格的篩選程序，刪除語音清晰度問題的查詢，具體排除任何在 Whisper Radford 等人 (2023) 假設中呈現非零詞錯誤率的發音。我們也刪除由 UTMOS (Saeki 等人，2022) 分數低於 3.5 確定的音質不佳的發音。這產生了一組高品質的 1,862 個語音查詢，我們稱之為 AudioCRAG-Synthetic 基準。

Audio CRAG Human: To further enhance the realism and diversity of our evaluation, we introduce the AudioCRAG-Human benchmark, which consists of 618 human-recorded spoken queries. These queries are recorded by a diverse pool of participants to capture natural variations in speech, accent, and prosody. The inclusion of human-recorded audio enables a more comprehensive assessment of speech-in speech-out systems under real-world conditions, providing valuable insights into model robustness and generalization beyond synthetic speech. This benchmark serves as a critical resource for evaluating the effectiveness of tool integration in conversational AI systems.

D Tool Usage Setup

To enable effective tool usage, we build upon the CRAG framework by integrating two complementary information sources: web search, which provides access to fresh and dynamic content, and a knowledge graph, which offers structured and reliable information. For web search, we aggregate all 100,000 documents from the CRAG corpus and employ a BGE-based re-ranker[2] Xiao et al. (2023) to index and retrieve the top 50 most relevant documents for each query. These documents are then segmented into chunks and re-ranked using the same BGE model based on their similarity to the query, ensuring highly contextually relevant retrieval. Meanwhile, queries to the knowledge graph are performed via a simulated API, adhering to the methodology established in CRAG [3].

Table 6 Results on AudioCRAG-Synthetic for Qwen2.5-7B, OpusLM, and Kimi Audio comparing text vs. speech output across Open Book setting showing average rates of accurate, hallucinated, and missing responses, as well as overall truthfulness scores for each system.

Ref length	Output	Model	Score	Acc.	Halluc	Miss.
0	Text	Qwen2.5-7B	-13.1	15.0	28.1	56.9
0	Speech	Qwen2.5-7B	-21.1	11.1	32.3	56.6
0	Text	OpusLM	-44.3	20.1	64.3	15.6
0	Speech	OpusLM	-47.9	18.4	66.2	15.4
0	Text	Kimi Audio	-38.5	24.2	62.7	13.1
0	Speech	Kimi Audio	-53.5	16.7	70.3	12.9

E Evaluation Setting

Similar to previous work (Yang et al., 2024), we employ model-based automatic evaluation. We use a three-way scoring system, assigning scores of 1, -1, and 0 for accurate, incorrect, and missing answers, respectively. The evaluation is conducted using the Llama 4-maverick LLM evaluator. For speech outputs, we first transcribe the audio using Whisper (Radford et al., 2023) before passing the transcriptions to the LLM evaluator. In this study, our primary focus is on enhancing system accuracy; therefore, we report average accuracy values in Tables 1 and 2. For additional context, we also provide the average rates of accurate, hallucinated, and missing responses, as well as overall truthfulness scores for each system in the Open Book Setting (see Table 6). Notably, our results indicate that Qwen-OMNI was less likely to generate hallucinated responses compared to OpusLM and Kimi Audio.

F Experiment Setup of SOTA speech-in speech-out models

Qwen-OMNI (Xu et al., 2025) is a end-to-end multimodal model that seamlessly integrates diverse input modalities—including text, images, audio, and video—and generates both text and natural speech responses in a real-time streaming fashion. It leverages an innovative Thinker-Talker architecture, where the Thinker

[2]https://huggingface.co/BAAI/bge-large-en-v1.5
[3]https://github.com/facebookresearch/CRAG/tree/main/mock_api

Audio CRAG Human：為了進一步增強評估的真實性和多樣性，我們引入了 AudioCRAG-Human 基準，其中包含 618 個人聲錄製的語音查詢。這些查詢由一群多樣化的參與者錄製，以捕捉語音、口音和音調的自然變化。人聲錄製音頻的包含使我們能夠在真實世界條件下對語音輸入語音輸出系統進行更全面的評估，提供有價值的見解，了解模型在合成語音之外的穩健性和泛化能力。這個基準作為評估工具整合在對話式 AI 系統中有效性的關鍵資源。

D 工具使用設定

為了實現有效的工具使用，我們基於 CRAG 框架，整合了兩種互補的資訊來源：網路搜尋，提供即時且動態的內容，以及知識圖，提供結構化且可靠的資訊。對於網路搜尋，我們匯集了 CRAG 資料庫中的全部 100,000 篇文檔，並使用基於 BGE 的重新排序器 [2] Xiao et al. (2023) 來索引並檢索每個查詢中最相關的前 50 篇文檔。這些文檔被分割成塊，並使用與查詢相似度相同的 BGE 模型重新排序，確保高度語境相關的檢索。同時，對知識圖的查詢是透過模擬 API 進行的，遵循 CRAG [3] 中建立的方 法論。

表 6 在 AudioCRAG-Synthetic 上針對 Qwen2.5-7B、OpusLM 和 Kimi Audio，在開放書籍設定下，比較文本與語音輸出的結果，顯示準確、虛構和缺失回應的平均率，以及每個系統的整體真實性分數。

參考長度	輸出	模型	分數	Acc.	虛構	錯誤
0	Text	Qwen2.5-7B	-13.1	15.0	28.1	56.9
0	語音	Qwen2.5-7B	-21.1	11.1	32.3	56.6
0	Text	OpusLM	-44.3	20.1	64.3	15.6
0	語音	OpusLM	-47.9	18.4	66.2	15.4
0	Text	Kimi Audio	-38.5	24.2	62.7	13.1
0	語音	Kimi Audio	-53.5	16.7	70.3	12.9

E 評估設定

類似先前研究 (Yang 等人，2024)，我們採用基於模型的自動評估。我們使用三向評分系統，分別為正確、錯誤和缺失答案分配 1、-1 和 0 分。評估使用 Llama 4-maverick LLM 評估器進行。對於語音輸出，我們首先使用 Whisper (Radford 等人，2023) 將音訊轉錄，然後將轉錄文本傳遞給 LLM 評估器。在本研究中，我們的主要重點是提升系統準確性；因此，我們在表 1 和表 2 中報告平均準確性值。為了提供更多背景資訊，我們也提供了準確、虛構和缺失回應的平均率，以及開放書籍設定中每個系統的整體真實性分數（見表 6）。值得注意的是，我們的結果顯示 Qwen-OMNI 相較於 OpusLM 和 Kimi Audio，較少產生虛構回應。

F SOTA 語音輸入語音輸出模型的實驗設置

Qwen-OMNI (Xu 等人，2025) 是一個端到端的多元模態模型，能無縫整合多種輸入模態——包括文本、圖像、音頻和視頻——並以實時流式方式生成文本和自然語音響應。它利用了一種創新的思考者-談話者架構，其中思考者模組進行高層級推理以產生文本響應，然後談話者模組基於文本以及思考者的隱藏表示進行條件化，以生成流式語音輸出。

[2]https://huggingface.co/BAAI/bge-large-en-v1.5[3]
https://github.com/facebookresearch/CRAG/tree/main/mock_api

module performs high-level reasoning to produce a text response, which is then used by the Talker module, conditioning on both the text and the Thinker's hidden representations, to generate streaming speech output.

OpusLM (Tian et al., 2025) is an open-source speech-in, speech-out model post-trained to directly answer complex semantic and factual questions from raw audio inputs, through Chain-of-Thought reasoning.

Kimi Audio (Ding et al., 2025) is a universal audio foundation model that unifies audio understanding, generation, and conversational abilities within a single framework. Pre-trained on over 13 million hours of diverse audio and text data, Kimi Audio achieves state-of-the-art performance across a wide range of audio benchmarks, including audio understanding and speech conversation tasks.

For tool-augmented scenarios, retrieval results are provided up to each model's maximum token limit ("Ref length" in Tables), maintaining a 2:1 ratio of web page to KG results. Specifically, we observe that Kimi-Audio is currently optimized for handling tool result references up to a certain length. When this limit is exceeded, an error arises during the audio detokenization process, specifically within the rotary embedding mechanism, highlighting an architectural constraint in processing longer input sequences or larger reference contexts. Addressing this limitation presents a valuable opportunity for future model enhancements.

Our evaluation encompasses both speech-in-text-out and speech-in-speech-out scenarios. For the Fixed-Interval Streaming RAG setting (Section 3.2.1), intermediate tool queries are generated at consistent 1-second intervals. In the Model-Triggered Streaming RAG setting, the model dynamically determines the need for a tool call after processing each 500ms block. This approach allows us to utilize a smaller chunk size, as only a single tool call thread is required for Model-Triggered Streaming RAG, thereby enabling more efficient and responsive processing.

G Post Training Data Preparation

This subsection details the experimental setup for post-training the pretrained speech-in, speech-out model to significantly enhance its tool usage capabilities, as outlined in S. 3.2.2. We leverage a random subset of 16,000 questions from the text-based factual question answering dataset TriviaQA (Joshi et al., 2017), which contains 97,000 questions and 662,659 associated web documents. For *Model-Triggered Streaming RAG*, we further compute word-level timestamps using a pre-trained ASR model, OWSM CTC v4 1B (Peng et al., 2025), enabling us to generate partial ASR transcripts X_b^{asr} at 500 ms intervals. Note that if word occurs at boundary of block b, it is excluded from X_b^{asr}. For each partial transcript X_b^{asr}, we generate corresponding psuedo ground truth queries $\overline{Q_b^T}$ using LLAMA-4-Maverick to simulate incremental user input. To simulate realistic tool usage, we concatenate all documents and employ a web query reranker to retrieve the top 50 most relevant documents for each query. The text questions from this 16k subset are converted into discrete speech tokens using text-to-speech synthesis with the corresponding pretrained speech-in, speech-out model. Recognizing that TriviaQA answers are typically single named entities, we further transform the queries into a conversational style using LLAMA-4-Maverick, making them more suitable for dialogue-based evaluation.

H Prompts used for Factual QA

H.1 Prompt in Closed Book Setting.

PROMPT = """ You are given an Audio Question and the time when it was asked in the Pacific Time Zone (PT), referred to as "Query Time". The query time is formatted as "mm/dd/yyyy, hh:mm:ss PT". Your task is to answer the question in as few words as possible.
Please follow these guidelines when formulating your answer:
1. If the question contains a false premise or assumption, answer "invalid question".
2. If you are uncertain or don't know the answer, respond with "I don't know".
Question
{query}
Query Time
{query_time}

模組執行高級推理以產生文本響應，然後談話者模組基於文本以及思考者的隱藏表示進行條件化，以生成流式語音輸出。

OpusLM (Tian 等人，2025) 是一個開源語音輸入、語音輸出的模型，經過後訓練，能夠直接從原始語音輸入中回答複雜的語義和事實問題，透過思考鏈推理。

Kimi Audio (Ding 等人，2025) 是一個通用語音基礎模型，將語音理解、生成和對話能力統一在單一框架內。在超過1300萬小時的多樣語音和文本數據上進行預訓練，Kimi Audio 在廣泛的語音基準測試中達到頂尖性能，包括語音理解和語音對話任務。

在工具增強場景中，檢索結果會提供到每個模型的最大標記限制（表格中的「Ref length」），維持網頁與知識庫結果的2:1比例。具體來說，我們觀察到 Kimi-Audio 目前已經優化處理工具結果參考的長度。當超出此限制時，在語音去標記化過程中會出現錯誤，具體是在旋轉嵌入機制中，突顯了處理更長輸入序列或更大參考上下文的架構限制。解決這個限制為未來模型的改進提供了寶貴的機會。

我們的評估涵蓋了語音輸入語音輸出和語音輸入語音輸出兩種場景。在固定間隔串流 RAG 設定（第 3.2.1 節）中，中間工具查詢會在一致的 1 秒間隔生成。在模型觸發串流 RAG 設定中，模型會在處理每個 500ms 块之後動態判斷是否需要工具呼叫。這種方法使我們可以使用更小的塊大小，因為模型觸發串流 RAG 只需要一個工具呼叫線程，從而實現更高效和反應更快的處理。

G 訓練後數據準備

本小節詳述了在預訓練的語音輸入語音輸出模型上進行後訓練的實驗設置，以顯著提升其工具使用能力，如 S. 3.2.2 所述。我們利用來自基於文本的事實性問答數據集 TriviaQA（Joshi 等人，2017）的隨機子集，該數據集包含 97,000 個問題和 662,659 個相關的網頁文檔。在模型觸發串流 RAG 中，我們使用預訓練的 ASR 模型 OWSM CTC v4 1B（Peng 等人，2025）進一步計算詞級時間戳，使我們能夠在 500 ms 間隔生成部分 ASR 詩篇 X_b^{asr}。請注意，如果詞語出現在塊 b 的邊界上，則會從 X_b^{asr}中排除。對於每個部分詩篇 X_b^{asr}，我們使用 LLAMA-4-Maverick 生成相應的虛假地面真實查詢 Q_b^T，以模擬增量用戶輸入。為了模擬真實的工具使用情況，我們將所有文檔連接起來，並使用網頁查詢重新排序器為每個查詢檢索前 50 個最相關的文檔。這個 16k 子集中的文本問題使用對應的預訓練語音輸入語音輸出模型通過文本到語音合成轉換為離散語音標記。認識到 TriviaQA 的答案通常是單個命名實體，我們使用 LLAMA-4-Maverick 將查詢轉換為對話風格，使其更適合基於對話的評估。

H Prompts 用於事實性問答

H.1 在閉本設定中的 Prompt

PROMPT = """ 你收到一個音頻問題以及問題提出的太平洋時區（PT）時間，稱為「查詢時間」。查詢時間的格式為「mm/dd/yyyy, hh:mm:ss PT」。你的任務是盡可能簡潔地回答問題。

請在組織答案時遵循以下指導原則：
1. 如果問題包含錯誤的前提或假設，請回答「無效問題」。
2. 若您不確定或不知道答案，請回應「我不知道」。
Question
{query}
Query Time
{query_time}

Table 7 Example KG Queries, and Web Queries generated by Qwen-OMNI in Open Book setting

ASR Transcript of Question X^{asr}	Web Query $\hat{Q}^{web}$	KG Query $\hat{Q}^{KG}$
which of nolan greenwald's movies has achieved the highest level of box office success on a global scale?	Nolan Greenwald's highest-grossing movie	{'domain': 'movie', 'movie_name': "Nolan Greenwald's movies", 'movie_aspect': 'revenue'}
who has played drums for the red hot chili peppers?	Red Hot Chili Peppers drummers	{'domain': 'music', 'artist_name': 'Red Hot Chili Peppers', 'artist_aspect': 'member'}
what's the current stock price of tortoise midstream energy fund?	Tortoise Midstream Energy Fund stock price	{'domain': 'finance', 'market_identifier': 'Tortoise Midstream Energy Fund', 'metric': 'price', 'datetime': '02/28/2024'}
what was the volume of trading in cabot corporation's stock on the most recent day that dividends were distributed?	CABOT Corp stock trading volume on dividend distribution date	{'domain': 'finance', 'market_identifier': 'Cabot Corporation', 'metric': 'dividend', 'datetime': '02/28/2024'}
which movie won the academy award for best film in 2020?	2020 Academy Award for Best Picture	{'domain': 'movie', 'movie_aspect': 'oscar_awards', 'year': 2020}
which teams have won against phoenix suns during 2022-12?	Teams that beat Phoenix Suns in December 2022	{'domain': 'sports', 'sport_type': 'basketball', 'team': 'Phoenix Suns', 'datetime': '2022-12-15'}

Answer
"""

H.2 Prompt in Open Book / Streaming RAG Setting.

PROMPT = """ You are given an Audio Question, References and the time when it was asked in the Pacific Time Zone (PT), referred to as "Query Time". The query time is formatted as "mm/dd/yyyy, hh:mm:ss PT". The references may or may not help answer the question. Your task is to answer the question in as few words as possible.
Please follow these guidelines when formulating your answer:
1. If the question contains a false premise or assumption, answer "invalid question".
2. If you are uncertain or don't know the answer, respond with "I don't know".
Question
{query}
Query Time
{query_time}
References
web
{web_results}
knowledge graph
{kg_response}
Answer
"""

表7 範例知識圖表示例查詢與網頁查詢，由Qwen-OMNI在開放書設置中生成

問題 X^{asr} 的語音轉文字記錄	網頁查詢 $\hat{Q}^{web}$	KG 查詢 $\hat{Q}^{KG}$
哪個 of 諾蘭 綠瓦爾德的電影是否已在全球規模上達到最高票房成績？	諾蘭·綠瓦爾德最高票房的電影 票房收入最高的電影	{'domain': 'movie', 'movie name': "Nolan" 格林瓦爾德的 電影", 'movie_長寬比': '收益'}
誰為紅色樂團打過鼓 辣椒？	Red Hot Chili Peppers 鼓手	{'domain': 'music', 'artist name': 'Red Hot Chili Peppers', 'artist_as- pect': '成員'}
目前的股票價格是 烏龜中游能源基金？	烏龜 中游 能源基金股價	{'domain': 'finance', 'mar- ket identifier': '烏龜中游能源基金', 'metric': 'price', 'datetime': '02/28/2024'}
CABOT Corp 股票在股息分發日期的最後一天交易量是多少？		{'domain': 'finance', 'mar- ket identifier': 'Cabot Corporation', 'metric': '股息' , 'datetime': '02/28/2024'}
哪部電影獲得了奧斯卡獎 2020年最佳電影獎？	2020 奧斯卡金像獎 最佳影片	{'domain': 'movie' 'movie_aspect': 'osc ar -_ 獎項', '年分：2020}
哪些球隊在 2022-12 期間擊敗了鳳凰城太陽隊？	擊敗鳳凰隊的球隊 2022 年 12 月	{'領域：'體育', '體育_類型：'籃球', '球隊：'鳳凰城太陽隊', 'datetime': '2022-12-15'}

回答「」

H.2 OpenBook /StreamingRAG 設定中的提示。

PROMPT = """ 您收到一個音訊問題、參考資料，以及問題提出時的太平洋時區（PT）時間，稱為「查詢時間」。查詢時間的格式為「mm/dd/yyyy, hh:mm:ss PT」。參考資料可能會或可能不會幫助回答問題。您的任務是盡可能簡潔地回答問題。

請在組織您的答案時遵循以下指南：
1. 如果問題包含錯誤的前提或假設，回答「無效問題」。
2. 如果您不確定或不知道答案，回答「我不知道」。
問題
{query} ### 查詢時間 {query_time} ## # 參考資料 # 網路 {web_results} # 知識圖表示例 {kg_response} ### 答案
"""

H.3 KG Query extraction in Open Book Setting.

PROMPT = """ You are an agent that only outputs JSON. You are given a Query and Query Time. Do the following:

1) Determine the domain the query is about. The domain should be one of the following: "finance", "sports", "music", "movie", "encyclopedia". If none of the domains apply, use "other". Use "domain" as the key in the result json.

2) Extract structured information from the query. Include different keys into the result json depending on the domains, and put them DIRECTLY in the result json. Here are the rules:

For 'encyclopedia' and 'other' queries, these are possible keys:
- 'main_entity': extract the main entity of the query.

For 'finance' queries, these are possible keys:
- 'market_identifier': stock identifiers including individual company names, stock symbols.
- 'metric': financial metrics that the query is asking about. This must be one of the following: 'price', 'dividend', 'P/E ratio', 'EPS', 'marketCap', and 'other'.
- 'datetime': time frame that the query asks about. When datetime is not explicitly mentioned, use 'Query Time' as default.

For 'movie' queries, these are possible keys:
- 'movie_name': name of the movie
- 'movie_aspect': if the query is about a movie, which movie aspect the query asks. This must be one of the following: 'budget', 'genres', 'original_language', 'original_title', 'release_date', 'revenue', 'title', 'cast', 'crew', 'rating', 'length'.
- 'person': person name related to moves
- 'person_aspect': if the query is about a person, which person aspect the query asks. This must be one of the following: 'acted_movies', 'directed_movies', 'oscar_awards', 'birthday'.
- 'year': if the query is about movies released in a specific year, extract the year

For 'music' queries, these are possible keys:
- 'artist_name': name of the artist
- 'artist_aspect': if the query is about an artist, extract the aspect of the artist. This must be one of the following: 'member', 'birth place', 'birth date', 'lifespan', 'artist work', 'grammy award count', 'grammy award date'.
- 'song_name': name of the song
- 'song_aspect': if the query is about a song, extract the aspect of the song. This must be one of the following: 'author', 'grammy award count', 'release country', 'release date'.

For 'sports' queries, these are possible keys:
- 'sport_type': one of 'basketball', 'soccer', 'other'
- 'tournament': NBA, World Cup, Olympic.
- 'team': teams that users are interested in.
- 'datetime': time frame that the user is interested in. When datetime is not explicitly mentioned, use 'Query Time' as default.

Return the results in a FLAT json.

NEVER include ANY EXPLANATION or NOTE in the output, ONLY OUTPUT JSON!!!
"""

H.3 KGQuery提取在开放書籍設定中。

PROMPT = """ 你是一個只輸出 JSON 的代理。你會得到一個查詢和查詢時間。請執行以下操作：

1) 判定查詢的領域。領域應該是以下其中之一："finance"、"sports"、"music"、"movie"、"encyclopedia"。若沒有任何領域適用，請使用"other"。在結果 json 中以"domain"作為鍵。

2) 從查詢中提取結構化信息。根據不同領域，將不同的鍵包含在結果 json 中，並直接將它們放入結果 json。這裡是規則：

對於『encyclopedia』和『other』查詢，這些是可能的鍵：
- 'main_entity'：提取查詢的主體。

對於『finance』查詢，這些是可能的鍵：
- 'market_identifier'：包含個別公司名稱、股票代碼的股票識別碼。
- 'metric'：查詢所關注的金融指標。這必須是以下其中之一：'價格'，'股息'，'市盈率'，'每股收益'，'市值'，和'其他'。
- '日期時間'：查詢所關注的時間範圍。當日期時間未明確提及時，使用'查詢時間'作為預設值。

對於'電影'查詢，這些是可能的鍵：
- 'movie_name：電影的名稱
- 'movie_aspect'：如果查詢是關於電影，查詢詢問的是電影的哪個方面。這必須是以下其中之一：'budget'，'genres'，'original_language'，'original_title'，'release_date'，'revenue'，'title'，'cast'，'crew'，'rating'，'length'。

- 'person'：與移動相關的人名
- 'person_aspect'：如果查詢是關於人，那麼查詢詢問的是哪個人的方面。這必須是以下其中之一：'acted_movies'，'directed_movies'，'oscar_awards'，'birthday'。
- '年分'：如果查詢是關於在某特定年分發行的電影，則提取該年分

針對「音樂」查詢，這些是可能的鍵：
-「artist_name」：藝術家的名字
-「artist_aspect」：如果查詢是關於藝術家，則提取藝術家的面向。這必須是以下其中之一：「成員」、「出生地」、「出生日期」、「生涯」、「藝術家作品」、「格萊美獎數量」、「格萊美獎日期」。

-「song_name」：歌曲的名字
- 'song_aspect'：如果查詢是關於一首歌，提取歌曲的方面。這必須是以下其中之一：'作者'，'格萊美獎獲獎次數'，'發行國家'，'發行日期'。

為了「運動」相關的查詢，這些是可能的關鍵字：
- '運動'_type：'籃球'、'足球'、'其他'
- '賽事'：NBA, 世界盃, 奧林匹克。
- '團隊'：使用者感興趣的團隊。
- 'datetime'：使用者感興趣的時間範圍。當 datetime 未明確提及時，預設使用「查詢時間」。

以 FLAT json 格式回傳結果。

絕對不包含任何解釋或註解在輸出中，僅輸出JSON！！！ """

H.4 KG Query extraction in Streaming RAG Setting.

PROMPT = """ You are an agent that only outputs JSON. You are given an Audio Query, Previously generated JSON result ('Previous Result') and Query Time. Do the following:

1) Determine the domain the query is about. The domain should be one of the following: f̈inance, s̈ports, m̈usic, m̈ovie, ëncyclopedia. If none of the domains apply, use öther. Use d̈omainäs the key in the result json.

2) Extract structured information from the query. Include different keys into the result json depending on the domains, and put them DIRECTLY in the result json. Here are the rules:

For 'encyclopedia' and 'other' queries, these are possible keys: - 'main_entity': extract the main entity of the query.

For 'finance' queries, these are possible keys: - 'market_identifier': stock identifiers including individual company names, stock symbols. - 'metric': financial metrics that the query is asking about. This must be one of the following: 'price', 'dividend', 'P/E ratio', 'EPS', 'marketCap', and 'other'. - 'datetime': time frame that the query asks about. When datetime is not explicitly mentioned, use 'Query Time' as default.

For 'movie' queries, these are possible keys: - 'movie_name': name of the movie - 'movie_aspect': if the query is about a movie, which movie aspect the query asks. This must be one of the following: 'budget', 'genres', 'original_language', 'original_title', 'release_date', 'revenue', 'title', 'cast', 'crew', 'rating', 'length'. - 'person': person name related to moves - 'person_aspect': if the query is about a person, which person aspect the query asks. This must be one of the following: 'acted_movies', 'directed_movies', 'oscar_awards', 'birthday'. - 'year': if the query is about movies released in a specific year, extract the year

For 'music' queries, these are possible keys: - 'artist_name': name of the artist - 'artist_aspect': if the query is about an artist, extract the aspect of the artist. This must be one of the following: 'member', 'birth place', 'birth date', 'lifespan', 'artist work', 'grammy award count', 'grammy award date'. - 'song_name': name of the song - 'song_aspect': if the query is about a song, extract the aspect of the song. This must be one of the following: 'author', 'grammy award count', 'release country', 'release date'.

For 'sports' queries, these are possible keys: - 'sport_type': one of 'basketball', 'soccer', 'other' - 'tournament': NBA, World Cup, Olympic. - 'team': teams that users are interested in. - 'datetime': time frame that the user is interested in. When datetime is not explicitly mentioned, use 'Query Time' as default. Return the results in a FLAT json.
NEVER include ANY EXPLANATION or NOTE in the output, ONLY OUTPUT JSON!!!

3) Compare your newly generated result to the 'Previous Result'. **If your new result would be exactly the same as the 'Previous Result', output only NO_QUERY.** Return the results in a FLAT json.

Previous Result:
{prev_kg_query}
"""

H.5 Web Query extraction in Open Book Setting.

PROMPT = """ You are given an Audio Query and Query Time. Your task is to generate a web query that can be used to retrieve relevant web pages. Rewrite the following query into a short and succinct form, focusing on the main topic or domain (e.g. finance, sports, music, movie, encyclopedia), key entities mentioned (e.g. people, organizations, locations), and specific aspects of those entities (e.g. performance metrics, relationships, events). Ensure the rewritten query is clear, concise, and easy to understand. Note that simply outputting the original query is not acceptable. You must rephrase the query to make it more concise and focused on the key information that will help retrieve relevant web pages.

For 'finance' queries, focus on: - Company names or stock symbols - Financial metrics (e.g. price, dividend, P/E ratio, EPS, marketCap) - Specific timeframes or events; if no timeframe is specified, use the Query Time as default

H.4 KGQueryextractionin StreamingRAG 設定。

PROMPT = """ 你是一個只輸出JSON的代理。你收到一個音訊查詢、先前生成的JSON結果（'先前結果'）和查詢時間。請執行以下操作：

1) 判定查詢的領域。該領域應為下列其中之一：f̈inance（財務）、s̈ports（運動）、m̈usic（音樂）、m̈ovie（電影）、ëncyclopedia（百科全書）。若無適用領域，則使用 öther。使用d̈omainäs 為結果 json 中的鍵值。

2) 從查詢中擷取結構化資訊。根據
網域，並將它們直接放入結果 JSON 中。這裡是規則：

針對「百科全書」和「其他」查詢，這些是可能的鍵：- 「main_entity」：提取查詢的主體。

針對「金融」查詢，這些是可能的鍵：- 「market_identifier」：股票識別碼，包括個別公司名稱和股票代碼。- 「metric」：查詢中涉及的金融指標。這必須是以下其中之一：「價格」、「股息」、「P/E 比率」、「EPS」、「市值」和「其他」。- 「datetime」：查詢涉及的時間範圍。當 datetime 未明確提及時，使用「查詢時間」作為預設值。

針對「電影」查詢，這些是可能的鍵：- 「movie_name」：電影的名稱- 「movie_aspect」：如果查詢是關於電影，則查詢涉及的電影面向。這必須是以下其中之一：「預算」、「類型」、「原始_語言」、「原始_標題」、「發行_日期」、「票房」、「標題」、「演員」、「劇組」、「評分」、「片長」。
- 'person'：關於移動相關的人名 - 'person_aspect'：如果查詢是關於人，則提取查詢所問的人的方面。這必須是以下其中之一：'acted_movies'，'directed_movies'，'oscar_awards'，'birthday'。- 'year'：如果查詢是關於特定年份發行的電影，則提取該年份

For 'music' 查詢，這些是可能的鍵：- 'artist_name'：藝術家的名字 - 'artist_aspect'：如果查詢是關於藝術家，則提取藝術家的方面。這必須是以下其中之一：'member'，'birth place'，'birth date'，'lifespan'，'artist work'，'grammy award count'，'grammy award date'。- 'song_name'：歌曲的名字 - 'song_aspect'：如果查詢是關於歌曲，則提取歌曲的方面。這必須是以下其中之一：'author'，'grammy award count'，'release country'，'release date'。

the song - 'song_aspect'：如果查詢是關於歌曲，則提取歌曲的方面。這必須是以下其中之一：'author'，'grammy award count'，'release country'，'release date'。

對於「運動」相關的查詢，這些是可能的關鍵字：- 'sport_type'：包含「籃球」、「足球」、「其他」其中一項 - 'tournament'：NBA、世界盃、奧林匹克 - 'team'：使用者感興趣的隊伍 - 'datetime'：使用者感興趣的時間範圍。當 datetime 未明確提及時，預設使用「查詢時間」。以扁平的 json 格式回傳結果。

永遠不包含任何解釋或註解，只輸出JSON！！！

3) 將您新產生的結果與「先前結果」進行比較。**如果您的結果會完全符合
與「上一個結果」相同，僅輸出 NO_QUERY。** 請以扁平化的 json 格式回傳結果。

Previous Result:
{prev_kg_query}
"""

H.5 網頁查詢在開放書籍設定中的萃取。

PROMPT = """ 您已提供音訊查詢與查詢時間。您的任務是產生一個可用於查詢相關網頁的網路查詢。請將以下查詢改寫為簡潔的形式，重點在於主題或領域（例如：財經、體育、音樂、電影、百科）、提及的關鍵實體（例如：人、組織、地點），以及這些實體的具體面向（例如：績效指標、關係、事件）。確保改寫後的查詢清晰、簡潔且容易理解。請注意，僅僅輸出原始查詢是不被接受的。您必須重新組織查詢，使其更簡潔並聚焦於有助於查詢相關網頁的關鍵資訊。

針對「財務」相關查詢，請重點關注：- 公司名稱或股票代碼 - 財務指標（例如：價格、股息、市盈率、每股收益、市值）- 單一特定時間範圍或事件；若未指定時間範圍，則以查詢時間為預設值

For 'sports' queries, focus on: - Sports Type (eg. basketball, soccer) - Teams, players - Statistics or performance metrics (e.g. scores, wins, losses) - Specific events or tournaments (eg. NBA, World Cup, Olympic) - Time frame that the user is interested in; if no timeframe is specified, use the Query Time as default

For 'music' queries, focus on: - Artist names or song titles - Specific aspects of artist (eg. band name, birth place, birth date, lifespan, artist work, grammy award count, grammy award date) - Specific aspects of song (eg. author, grammy award count, release country, release date) - Music genres or categories - Specific awards or recognition (e.g. Grammy Awards, Billboard)

For 'movie' queries, focus on: - Movie titles or celebrity names - Movie genres or other categories like budget, language, release_date, revenue, cast, crew, rating, length - Specific aspects of celebrity like acted_movies, directed_movies, oscar_awards, birthday - Specific awards or recognition (e.g. Oscars) For 'other' queries, focus on: - Main entity or topic - Specific aspects or attributes of the entity

When rewriting the query, ensure that it captures all important information from the original question that could impact the retrieval results. Do not omit any crucial details, such as specific dates, locations, or relationships between entities. Also, do not invent any new details on your own. If necessary, use the Query Time to provide context for the query. The goal is to create a concise and accurate query that effectively conveys the user's intent and retrieves relevant information. *NEVER include ANY EXPLANATION or NOTE in the output, ONLY OUTPUT THE REWRITTEN QUERY!!!* """

H.6 Web Query extraction in Streaming RAG Setting.

PROMPT = """You are given an Audio Query, previously generated Web query ('Previous Result') and Query Time.

Your task is to generate a web query that can be used to retrieve relevant web pages. Rewrite the following query into a short and succinct form, focusing on the main topic or domain (e.g. finance, sports, music, movie, encyclopedia), key entities mentioned (e.g. people, organizations, locations), and specific aspects of those entities (e.g. performance metrics, relationships, events). Ensure the rewritten query is clear, concise, and easy to understand.

Note that simply outputting the original query is not acceptable. You must rephrase the query to make it more concise and focused on the key information that will help retrieve relevant web pages.

For 'finance' queries, focus on: - Company names or stock symbols - Financial metrics (e.g. price, dividend, P/E ratio, EPS, marketCap) - Specific timeframes or events; if no timeframe is specified, use the Query Time as default

For 'sports' queries, focus on: - Sports Type (eg. basketball, soccer) - Teams, players - Statistics or performance metrics (e.g. scores, wins, losses) - Specific events or tournaments (eg. NBA, World Cup, Olympic) - Time frame that the user is interested in; if no timeframe is specified, use the Query Time as default

For 'music' queries, focus on: - Artist names or song titles - Specific aspects of artist (eg. band name, birth place, birth date, lifespan, artist work, grammy award count, grammy award date) - Specific aspects of song (eg. author, grammy award count, release country, release date) - Music genres or categories - Specific awards or recognition (e.g. Grammy Awards, Billboard)

For 'movie' queries, focus on: - Movie titles or celebrity names - Movie genres or other categories like budget, language, release_date, revenue, cast, crew, rating, length - Specific aspects of celebrity like acted_movies, directed_movies, oscar_awards, birthday - Specific awards or recognition (e.g. Oscars)

For 'other' queries, focus on: - Main entity or topic - Specific aspects or attributes of the entity

When rewriting the query, ensure that it captures all important information from the original question that could impact the retrieval results. Do not omit any crucial details, such as specific dates, locations, or relationships between entities. Also, do not invent any new details on your own. If necessary, use the Query Time to provide context for the query. The goal is to create a concise and accurate query that effectively conveys the user's intent and retrieves relevant information. Now, compare the new web query to the previously generated web query ('Previous Result').

針對「體育」相關查詢，請重點關注：- 體育類型（例如：籃球、足球）- 球隊、球員 - 統計數據或表現指標（例如：比賽分數、勝場、敗場）- 單一特定賽事或錦標賽（例如：NBA、世界盃、奧運會）- 使用者感興趣的時間範圍；若未指定時間範圍，則以查詢時間為預設值
針對「音樂」查詢，重點關注：- 藝術家名稱或歌曲名稱 - 藝術家特定方面（例如：樂團名稱、出生地、出生日期、生涯、藝術家作品、格萊美獎獲獎次數、格萊美獎獲獎日期）- 歌曲特定方面（例如：作者、格萊美獎獲獎次數、發行國家、發行日期）- 音樂流派或分類 - 特定獎項或認可（例如：格萊美獎、Billboard）

針對「電影」查詢，重點關注：- 電影名稱或名人名稱 - 電影流派或其他分類，如預算、語言、發行_日期、票房、演員、團隊、評分、片長 - 名人特定方面，如演過_的電影、導過_的電影、奧斯卡_獎、生日 - 特定獎項或認可（例如：奧斯卡獎）針對「其他」查詢，重點關注：- 主要實體或主題 - 實體的特定方面或屬性

在重寫查詢時，確保它捕捉到原始問題中所有可能影響檢索結果的重要資訊。不要省略任何關鍵細節，例如特定日期、地點或實體之間的關係。此外，也別自行捏造任何新細節。如有必要，使用查詢時間提供查詢的背景資訊。目標是創建一個簡潔且準確的查詢，有效地傳達用戶的意圖並檢索相關資訊。*絕不包含任何解釋或註記在輸出中，僅輸出重寫後的查詢！！！* """

H.6 StreamingRAG設定中的網頁查詢萃取。

PROMPT = """您已經收到一個音訊查詢、先前生成的網頁查詢（『先前結果』）以及查詢時間。

您的任務是生成一個可用於擷取相關網頁的網頁查詢。將以下查詢改寫為簡潔的形式，重點放在主要主題或領域（例如：財務、體育、音樂、電影、百科全書）、提及的關鍵實體（例如：人、組織、地點），以及這些實體的特定面向（例如：績效指標、關係、事件）。確保改寫後的查詢清晰、簡潔且容易理解。

請注意，僅僅輸出原始查詢是不被接受的。您必須重新組織查詢，使其更簡潔並聚焦於有助於擷取相關網頁的關鍵資訊。

對於『財務』查詢，重點放在：- 公司名稱或股票代碼- 財務指標（例如：價格、股息、市盈率、每股收益、市值）- 特定時間範圍或事件；如果未指定時間範圍，則使用查詢時間作為預設值

針對「運動」相關查詢，重點關注：- 運動種類（例如籃球、足球）- 球隊、球員 - 統計數據或表現指標（例如比數、勝場、敗場）- 單一事件或賽事（例如NBA、世界盃、奧運會）- 使用者感興趣的時間範圍；若未指定時間範圍，則以查詢時間為預設值

針對「音樂」相關查詢，重點關注：- 藝術家名稱或歌曲名稱 - 藝術家特定方面（例如樂團名稱、出生地、出生日期、生涯、藝術家作品、格萊美獎數量、格萊美獎日期）- 歌曲特定方面（例如作者、格萊美獎數量、發行國家、發行日期）- 音樂流派或類別 - 特定獎項或認可（例如格萊美獎、Billboard）

針對「電影」相關查詢，重點關注：- 電影名稱或名人名稱 - 電影類型或其他分類（例如預算、語言、發行_日期、票房、演員、劇組、評分、片長）- 名人特定方面（例如演過_的電影、導過_的電影、奧斯卡_獎項、生日）- 特定獎項或認可（例如奧斯卡獎）

針對「其他」詢問，重點在於：- 主要實體或主題 - 實體的具體面向或屬性

當重寫查詢時，確保它能捕捉到原始問題中所有重要的資訊，這些資訊可能會影響檢索結果。請勿省略任何關鍵細節，例如特定日期、地點或實體之間的關係。此外，也請勿自行編造任何新的細節。如有必要，可以使用「查詢時間」來提供查詢的背景資訊。目標是創建一個簡潔且準確的查詢，能夠有效地傳達用戶的意圖並檢索相關資訊。現在，請將新的網路查詢與先前生成的網路查詢（「先前結果」）進行比較。

If the new query is similar enough to the previous web query (i.e., it effectively conveys the same user intent and would retrieve similar relevant information), output only *NO_QUERY*.

Previous Result:
{prev_web_query}
"""

I LLM as judge prompt

PROMPT="""
Assume you are a human expert in grading predictions given by a model. You are given a question and a model prediction. Judge if the prediction matches the ground truth answer by following these steps: 1: Take it as granted that the Ground Truth is always correct. 2: If the Prediction indicates it is not sure about the answer, "score" should be "0"; otherwise, go the next step. 3: If the Prediction exactly matches the Ground Truth, "score" is 1. 4: If the Prediction does not exactly match the Ground Truth, go through the following steps and likely give a score as 0. 5: If the Ground Truth is a number, "score" is 1 if and only if the Prediction gives a number that almost exactly matches the ground truth. 6: If the Prediction is self-contradictory, "score" must be 0. 7: If the prediction is not answering the question, "score" must be 0. 8: If the prediction is a concise and correct summary of the ground truth, "score" is 1. 9: If ground truth contains a set of items, prediction must contain exactly same items for the score to be 1. 10: Otherwise, "score" is 0.

Output a JSON blob with an "explanation" field explaining your answer as short as possible and an "score" field with value 1 or 0.

You should make the judgment based on provided examples. Examples: Question: "which company has higher eps, btu or cma?" Ground Truth: "cma" Prediction: "it is not possible to determine which company has a higher eps." Output: "score": 0, "explanation": "The prediction is not sure about the answer."

Question: "who is taller, a or b?" Ground Truth: "a" Prediction: "The answer is a. a is of height 1.75 m and b is of height 1.82 m. So b is taller." Output: "score": 0, "explanation": "The prediction is self-contradictory."

Question: "who among patricia place and lana clarkson is younger?" Ground Truth: "lana clarkson" Prediction: "lana clarkson (born april 5, 1962) is younger than patricia clarkson (born december 29, 1959)." Output: "score": 1, "explanation": "The prediction gives the correct answer."

Question: "did lakers win more games than warriors in 2024-4?" Ground Truth: "yes" Prediction: "yes, lakers won more games than warriors in 2023-2024 season." Output: "score": 0, "explanation": "The prediction is not answering the question."

Question: "what is the height of the building where citigroup is headquartered?" Ground Truth: "151 m" Prediction: "the height is 915 feet (279 m)." Output: "score": 0, "explanation": "The prediction, 151 m, does not match the ground truth, 279 m."

Question: "who were the last 3 prime ministers of uk?" Ground Truth: "rishi sunak, liz truss, and boris johnson" Prediction: "1. liz truss, 2. boris johnson, 3.theresa may" Output: "score": 0, "explanation": "The prediction does not contain item, rishi sunak, that is in the grount truth."

Question: "who authored the taming of the shrew?" Ground Truth: "william shakespeare" Prediction: "william shakespeare" Output: "score": 1, "explanation": "The prediction exactly matches the ground truth."

Question: "who authored The Taming of the Shrew?" Ground Truth: "william shakespeare" Prediction: "w shakespeare" Output: "score": 1, "explanation": "The prediction matches the ground truth as w is the abbreviation of william."

Question: "what is the state bird of california?" Ground Truth: "california quail" Prediction: "california valley quail" Output: "score": 1, "explanation": "The prediction matches the ground truth as california quail is also known as california valley quail."

如果新的查詢與先前的網路查詢相似 enough（即，它能夠有效地傳達相同的用戶意圖，並且會檢索相似的相關資訊），請輸出*NO_QUERY*。

先前結果：
{prev_web_query}
"""

我 LLM 作為評判提示

PROMPT="""
假設你是一位精通評分模型預測的專家。你將獲得一個問題和模型的預測。請按照以下步驟判斷預測是否與真實答案相符：1：假設真實答案總是正確的。2：如果預測表示對答案不確定，"score" 應該是 "0"；否則，進行下一步。3：如果預測與真實答案完全一致，"score" 是 1。4：如果預測與真實答案不完全一致，請通過以下步驟進行評分，很可能給 0 分。5：如果真實答案是數字，"score" 是 1 的條件是預測給出了一個幾乎與真實答案匹配的數字。6：如果預測內部存在矛盾，"score" 必須是 0。7：如果預測沒有回答問題，"score" 必須是 0。8：如果預測是對真實答案的簡潔且正確的總結，"score" 是 1。9：如果真實答案包含一組項目，預測必須包含完全相同的項目，"score" 才是 1。10：否則，"score" 是 0。

輸出一個 JSON 物件，其中包含一個 "explanation" 字段來簡潔說明你的答案，以及一個 "score" 字段，其值為 1 或 0。

你應該根據提供的範例進行判斷。範例：問題："哪家公司每股收益更高，btu 或 cma？" 真實答案："cma" 預測："無法判斷哪家公司每股收益更高." 輸出："score": 0, "explanation": "預測對答案沒有把握。"

問題："誰更高，a 或 b？" 真實答案："a" 預測："答案是 a。a 的身高是 1.75 公尺，b 的身高是 1.82 公尺。所以 b 更高。" 輸出："score": 0, "explanation": "預測內容自相矛盾。"

問題："patricia place 和 lana clarkson 中哪一位年輕？" 真實答案："lana clarkson" 預測："lana clarkson（1962年4月5日出生）比 patricia clarkson（1959年12月29日出生）年輕。" 輸出："score": 1, "explanation": "預測提供了正確的答案。"

問題：『2024-4季節，湖人隊贏的比勇士隊多嗎？』事實：『是』預測：『是，湖人隊在2023-2024賽季贏的比勇士隊多。』輸出：『分數』：0，『解釋』：『預測沒有回答問題。』

問題：『花旗集團總部大樓的高度是多少？』事實：『151公尺』預測：『高度是915英尺（279公尺）。』輸出：『分數』：0，『解釋』：『預測，151公尺，與事實，279公尺，不符。』

問題：『英國最後三位首相是誰？』事實：『里希·蘇納克、莉茲·特拉斯和喬治·貝納森』預測：『1. 莉茲·特拉斯，2. 喬治·貝納森，3. 特蕾莎·梅』輸出：『分數』：0，『解釋』：『預測未包含事實中提到的項目，里希·蘇納克。』

問題：『誰是《驯悍记》的作者？』事實：『威廉·莎士比亞』預測：『威廉·莎士比亞』輸出：『分數』：1，『解釋』：『預測與事實完全一致。』

問題：『誰是《驯悍记》的作者？』事實：『威廉·莎士比亞』預測：『w 莎士比亞』輸出：『分數』：1，『解釋』：『預測與事實相符，因為w是威廉的縮寫。』

問題：『加州的州鳥是什麼？』 真實答案：『加州雉』 預測：『加州山谷雉』 輸出：『分數』：1，『解釋』：『預測與真實答案相符，因為加州雉也稱為加州山谷雉。』

Question: "what country is maharashtra metro rail corporation limited located in?" Ground truth: "india" Prediction: "indianapolis" Output: "score": 0, "explanation": "The prediction doesn't match the ground truth answers."

Question: "what was wolves's record in their last game in eng-premier league?" Ground Truth: "w" Prediction: "wolves's record in their last game in the eng-premier League was a 2-1 win over fulham." Output: "score": 1, "explanation": "The prediction correctly answers the result is a win."

Question: "how many seconds is 3 minutes 15 seconds?" Ground Truth: "195 seconds" Prediction: "3 minutes 15 seconds is 195 seconds." Output: "score": 1, "explanation": "The prediction matches ground truth as it correctly states the number of seconds asked in the question."

Question: "on which days did am distribute dividends in the last year?" Ground Truth: "2023-01-24, 2023-04-25, 2023-07-25, 2023-10-24" Prediction: "the company distributed dividends on october 24, 2023 and january 24, 2023." Output: "score": 0, "explanation": "The prediction partially matches the ground truth, but it is incomplete."

Question: "on which days did xxx distribute dividends in the last year?" Ground Truth: "2023-01-13, 2023-03-25, 2023-11-21" Prediction: "xxx distributed dividends on 1. 2023-01-13, 2. 2023-03-25, 3. 2023-10-21." Output: "score": 0, "explanation": "The prediction contains an item, 2023-10-21, which is not in ground truth."

Question: "who are current band members of maroon 5?" Ground Truth: "adam Levine, jesse carmichael, james valentine, matt flynn, pj morton, sam farrar" Prediction: "they are jesse carmichael, matt flynn, adam levine, pj morton, sam farrar, and james valentine." Output: "score": 1, "explanation": "The prediction exactly matches the ground truth."

Question: "which movies comprise the matrix franchise?" Ground Truth: "the matrix, the matrix reloaded, the matrix revolutions, the matrix resurrections" Prediction: "the matrix, the matrix reloaded, the matrix revolutions, the animatrix, and the matrix resurrections." Output: "score": 0, "explanation": "The prediction covers more items than what are given by the ground truth."

Question: "how deep is the deepest lake of new york?" Ground Truth: "618 ft" Prediction: "the deepest lake in new york is seneca lake, with a depth of 618.23 feet." Output: "score": 1, "explanation": "The prediction exactly matches the number in ground truth after rounding."

Question: "what is the closing price of meta yesterday?" Ground Truth: "$310.17" Prediction: "310.2" Output: "score": 1, "explanation": "The prediction exactly matches the number in ground truth after rounding."

Question: "what is the current market cap of appl?" Ground Truth: "2.81 trillion" Prediction: "2.667 trillion" Output: "score": 0, "explanation": "The prediction does not match the number in ground truth."

Question: "what is the current pe ratio of appl?" Ground Truth: "28.3" Prediction: "the current pe ratio of apple is 26.66" Output: "score": 0, "explanation": "The prediction does not match the number in ground truth."

Question: "how much is tesla's stock price down from its all-time high?" Ground Truth: "$221.83" Prediction: "209.52" Output: "score": 0, "explanation": "The prediction does not match the number in ground truth."

Question: "what is the length of amazon river?" Ground Truth: "over 4000 miles" Prediction: "the length of amazon river is 4,000 miles" Output: "score": 0, "explanation": "The prediction does not say Amazon River is longer than 4000 miles."

Question: "how many copies x were sold?" Ground Truth: "2 million." Prediction: "it is over 2 million" Output: "score": 0, "explanation": "The prediction does not match the ground truth."

Question: "what is the population of country x?" Ground Truth: "3,576,873" Prediction: "the population of country x is 3.3 million." Output: "score": 0, "explanation": "The prediction, 3.3 M, does not match the number, 3.6 M, in ground truth."

Question: "what is the current market value of stock x?" Ground Truth: "$2,237,578,268" Prediction: "$2.16 billion." Output: "score": 0, "explanation": "The prediction, 2.16 B, does not match the number, 2.2 B, in

問題：『馬哈拉施特拉地鐵公司有限責任公司位於哪個國家？』 真實答案：『印度』 預測：『印第安那波利斯』 輸出：『分數』：0，『解釋』：『預測與真實答案不符。』

問題：『狼隊在英格蘭超級聯賽最後一場比賽的成績是什麼？』事實：『w』預測：『狼隊在英格蘭超級聯賽最後一場比賽的成績是以2-1擊敗富咸。』輸出：『分數』：1，『解釋』：『預測正確回答了結果是勝利。』

問題：『3分15秒是多少秒？』事實：『195秒』預測：『3分15秒是195秒。』輸出：『分數』：1，『解釋』：『預測與事實相符，因為它正確說明了問題中問的秒數。』

問題：『在過去一年中，安公司分紅的日期是哪幾天？』事實：『2023-01-24，2023-04-25，2023-07-25，2023-10-24』預測：『公司在2023年10月24日和2023年1月24日分紅。』輸出：『分數』：0，『解釋』：『預測部分符合事實，但它不完整。』

問題：『xxx 在過去一年中哪些日期發放了股息？』事實：『2023-01-13, 2023- 03-25, 2023-11-21』預測：『xxx 於 1. 2023-01-13, 2. 2023-03-25, 3. 2023-10-21 發放了股息。』輸出：『分數』：0, 『解釋』：『預測包含一項，2023-10-21，這項不在事實中。』

問題：『maroon 5 的現任成員有哪些？』事實：『adam Levine, jesse carmichael, james valentine, matt flynn, pj morton, sam farrar』預測：『他們是 jesse carmichael, matt flynn, adam levine, pj morton, sam farrar, 與 james valentine。』輸出：『分數』：1, 『解釋』：『預測完全符合事實。』

問題：『哪些電影組成了「黑客帝国」系列？』真實答案：『黑客帝国、黑客帝国重装上阵、黑客帝国革命、黑客帝国復活』預測：『黑客帝国、黑客帝国重装上阵、黑客帝国革命、终极矩阵，以及黑客帝国復活。』輸出：『分數』：0、『解釋』：『預測涵蓋了比真實答案提供的更多項目。』

問題：『紐約最深湖的深度是多少？』真實答案：『618英尺』預測：『紐約最深湖是塞尼卡湖，深度為618.23英尺。』輸出：『分數』：1、『解釋』：『預測與真實答案中的數字在四捨五入後完全匹配。』

問題：『Meta昨日的收盤價是多少？』真實答案：『310.17美元』預測：『310.2』輸出：『分數』：1、『解釋』：『預測與真實答案中的數字在四捨五入後完全匹配。』

問題：『appl 的目前市值是多少？』實際答案：『2.81 兆』預測：『2.667 兆』輸出：『score』：0，『解釋』：『預測與實際答案中的數字不符。』

問題：『appl 的目前市盈率是多少？』實際答案：『28.3』預測：『apple 的目前市盈率是 26.66』輸出：『score』：0，『解釋』：『預測與實際答案中的數字不符。』

問題：『特斯拉的股價從創下最高點下跌了多少？』實際答案：『221.83 美元』預測：『209.52』輸出：『score』：0，『解釋』：『預測與實際答案中的數字不符。』

問題：『亞馬遜河的長度是多少？』實際答案：『超過 4000 英里』預測：『亞馬遜河的長度是 4,000 英里』輸出：『score』：0，『解釋』：『預測沒有說明亞馬遜河長度超過 4000 英里。』

問題：『x有多少份售出？』事實：『兩百萬份。』預測：『超過兩百萬份』輸出：『分數』：0，『解釋』：『預測與事實不符。』

問題：『x國的人口是多少？』事實：『3,576,873人』預測：『x國的人口是330萬人。』輸出：『分數』：0，『解釋』：『預測的330萬，與事實中的360萬不符。』

問題：『股票 x 的當前市場價值是多少？』事實：『$2,237,578,268』預測：『$2.16 億。』輸出：『分數』：0，『解釋』：『預測的 2.16 B 與數字 2.2 B 不符，

ground truth."

Question: "how old is michael jordan?" Ground Truth: "61" Prediction: "mj was born on February 17, 1963." Output: "score": 0, "explanation": "The prediction does not explicitly say the age."

Question: "how long did amy teach as a teacher?" Ground Truth: "24 years and 255 days" Prediction: "25 years." Output: "score": 0, "explanation": "The prediction does not exactly match the ground truth."

Question: "at what age did usher perform in the half time show of superbowl?" Ground Truth: "45 years old" Prediction: "usher has not performed at a super bowl halftime show." Output: "score": 0, "explanation": "The prediction does not match the ground truth."

Question: "what year did olivia rodrigo win the grammys best new artist award?" Ground Truth: "2022" Prediction: "sure! here's an answer in three words or fewer: 2022 grammy awards." Output: "score": 1, "explanation": "The prediction answers the correct number of the year."

Question: "how many times have zendaya and tom holland starred in a film together?" Ground Truth: "3" Prediction: "they have starred in 3 films together: spider-man: homecoming, spider-man: far from home, and spider-man: no way home." Output: "score": 1, "explanation": "The prediction gives the right number and it is reasonable."

Question: "who is younger, timothee chalamet or tom holland?" Ground Truth: "tom holland" Prediction: "timothée chalamet is younger than tom holland." Output: "score": 0, "explanation": "The prediction does not match the ground truth."

Question: "who had more number one hits on the us billboard, a or b?" Ground Truth: "a had more number one hits on the us billboard than b, with 20 number one hits compared to b's 15." Prediction: "a" Output: "score": 1, "explanation": "The prediction is a concise and correct summary of the ground truth."

Question: "what is xxx's birthdate?" Ground Truth: "1996-01-01." Prediction: "02/01/1996" Output: "score": 0, "explanation": "The prediction does not match the ground truth."

Question: "what was the worldwide box office haul for movie x?" Ground Truth: "101756123." Prediction: "102 million" Output: "score": 1, "explanation": "The prediction exactly matches the number in ground truth after rounding."

Question: "how much has spotify's user base increased by since 2020 in na?" Ground Truth: "spotify's user base increased by 34 million since 2020." Prediction: "spotify's north american user base increased from 36 million in 2020 to 85 million by 2021" Output: "score": 0, "explanation": "The prediction is not answering the question as it only gives the increase from 2020 to 2021."
"""

真實標準。

問題：『麥可·喬丹多大了？』事實：『61』預測：『mj 於 1963 年 2 月 17 日出生。』輸出：『分數』：0，『解釋』：『預測沒有明確說出年齡。』

問題：『艾米當老師有多久了？』真實標準：『24年又255天』預測：『25年。』輸出：『分數』：0，『解釋』：『預測並未精確符合真實標準。』

問題：『烏斯HER在超級碗半場秀中演出時幾歲？』真實標準：『45歲』預測：『烏斯HER從未在超級碗半場秀中演出過。』輸出：『分數』：0，『解釋』：『預測與真實標準不符。』

問題：『奧利維亞·羅德里戈獲得格萊美最佳新人獎是哪一年？』真實標準：『2022年』預測：『當然！這是一個三個字或更少的答案：2022年格萊美獎。』輸出：『分數』：1，『解釋』：『預測答對了年份的正確數字。』

問題：『澤丹亞和湯姆·荷蘭一起演過幾次電影？』真實標準：『3』預測：『他們一起演過3部電影：蜘蛛人：家鄉歸來、蜘蛛人：遠從家鄉、蜘蛛人：無法回家。』輸出：『分數』：1，『解釋』：『預測給出了正確的數字，且合理。』

問題：『蒂莫西·夏勒梅或湯姆·荷蘭誰年輕？』真實答案：『湯姆·荷蘭』預測：『蒂莫西·夏勒梅比湯姆·荷蘭年輕。』輸出：『分數』：0，『解釋』：『預測與真實答案不符。』

問題：『a 或 b 哪個在美國Billboard上擁有更多冠軍單曲？』真實答案：『a 在美國Billboard上擁有比b更多的冠軍單曲，a有20首冠軍單曲，而b有15首。』預測：『a』輸出：『分數』：1，『解釋』：『預測是簡潔且正確的概括真實答案。』

問題：『xxx的出生日期是什麼？』真實答案：『1996-01-01。』預測：『02/01/1996』輸出：『分數』：0，『解釋』：『預測與真實答案不符。』

問題：『電影x的全球票房收入是多少？』真實答案：『101756123。』預測：『1.02億』輸出：『分數』：1，『解釋』：『預測在四捨五入後與真實答案中的數字完全匹配。』

問題：『自2020年起，Spotify用戶基礎增加了多少？』事實真相：『自2020年起，Spotify的用戶基礎增加了3400萬。』預測：『Spotify的北美用戶基礎從2020年的3600萬增長至2021年的8500萬』輸出：『分數』：0，『解釋』：『預測未回應問題，僅提供了2020年至2021年的增長數據。』『』』

Table 8 Streaming Tool Queries generated by Qwen-OMNI after *Model-Triggered Streaming RAG* post-training

ASR Transcript of Question X^{asr}	**Partial ASR Transcript** X_b^{asr}	**Web Query** $\hat{Q}_b^{\text{web}}$	**KG Query** $\hat{Q}_b^{\text{KG}}$
Who founded Rare Beauty in 2019?	Who founded	Who founded what	{"domain": "other", "main_-entity": "Who"}
	Who founded rare	Red Bull founder	{"domain": "other", "main_-entity": "Red"}
	Who founded rare beauty	Who founded Rare Beauty	{"domain": "other", "main_-entity": "Rare Beauty"}
	Who founded rare beauty in	NO_QUERY	NO_QUERY
	Who founded rare beauty in 2019?	NO_QUERY	NO_QUERY
How many jump shots did Darius Miles score with in the game on 2000-11-08?	how many	Number of queries on 03/27/2024	{"domain": "other", "main_-entity": "How many"}
	how many jump	Number of jump shots in basketball	{"domain": "other", "main_-entity": "jump shot"}
	how many jump shots did	Derek Jeter jump shots count	{"domain": "sports", "sport_type": "other", "person": "Dairy", "date-time": "03/27/2024, 19:52:43 PT"}
	how many jump shots did Darius miles	Darius jump shots count	{"domain": "sports", "sport_type": "other", "person": "Darius", "date-time": "03/27/2024, 19:52:43 PT"}
	how many jump shots did Darius miles score	Darius Miles jump shots count	{"domain": "sports", "sport_type": "other", "person": "Darius Miles", "datetime": "03/27/2024, 19:52:43 PT"}
	how many jump shots did Darius miles score with in	Darius Miles jump shots scored	NO_QUERY
	how many jump shots did Darius miles score with in the game on	Darius Miles jump shots scored in game on 03/27/2024	NO_QUERY
	how many jump shots did Darius miles score with in the game on November	Darius Miles jump shots scored in game on November 2024	{"domain": "sports", "sport_type": "other", "person": "Darius Miles", "datetime": "November"}
	how many jump shots did Darius miles score with in the game on November 8	Darius Miles jump shots scored on November 8	{"domain": "sports", "sport_type": "other", "person": "Darius Miles", "datetime": "November 8"}
	how many jump shots did Darius miles score with in the game on November 8	NO_QUERY	NO_QUERY
	how many jump shots did Darius miles score with in the game on November 8, 2000	Darius Miles jump shots scored on November 8, 2000	{"domain": "sports", "sport_type": "other", "person": "Darius Miles", "datetime": "November 8, 2000"}

Table8 流式工具查詢，由 Qwen-OMNI 在模型觸發流式 RAG 訓練後生成

ASR Transcript of Question X^{asr}	**部分 ASR 語音字幕** X_b^{asr}	**Web 查詢** $\hat{Q}_b^{\text{web}}$	**KG 查詢** $\hat{Q}_b^{\text{KG}}$
誰在 2019 年創立了 Rare Beauty？	誰創立了	誰創立了什麼	{"domain": "other", "main_-entity": "誰"}
	誰創立了 rare	Red Bull創辦人	{"domain": "其他", "main_-entity": "紅"}
	誰創辦了Rare Beauty	Who founded Rare Beauty	{"domain": "其他","main_- 實體": "稀有美麗"}
	Who founded rare beauty in 誰在2019年創立了Rare Beauty？	NO_查詢	NO_查詢
		NO_查詢	NO_查詢
Darius Miles在 2000-11-08 那場比賽中投中了多少次跳投？	有多少	數量 查詢數量 03/27/2024 籃球跳投次數為- 籃球	{"domain": "other","main_-entity": "有多少"}
	有多少跳躍		{"domain": "other","主要_實體": "跳投"}
	Derek Jeter跳投次數為	Derek Jeter跳投次數統計	{"domain": "sports", "運動_類型": "其他", "人物": "Dairy", "日期時間": "03/27/2024,19:52:43 PT"}
	多少次跳投是達- ius miles	Darius 投籃得分	{"domain": "sports", "sport_type": "其他", "person": "Darius", "date- time": "03/27/2024,19:52:43 PT"}
	多少次跳投是達- iu士兵得分	Darius Miles 的跳投數量	{"domain": "sports", "sport_type": "其他", "person": "Darius Miles", "datetime": "03/27/2024, 19:52:43 PT"}
	Darius Miles 在此賽季有多少次跳投得分	Darius Miles 的跳投得分	NO_QUERY
	howmany跳投 shots didDarius miles score with in the game on	Darius Miles跳投在2024年3月27日的比賽中得分	NO_QUERY
	Darius Miles 在十一月比賽中投進了多少次跳投？ 十一月比賽	Darius Miles跳投在十一月比賽中投進 跳投 2024	{"domain": "運動 ts", "運動_類型": "其他 er", "人名": "達里厄斯·米爾 es", "日期時間": "十一月"}
	達里厄斯·米爾在十一月八日的比賽中投中了多少次跳投	達里厄斯 Miles跳躍於11月8日得分	{"domain": "sports", "sport_type": "其他", "person": "Darius Miles", "datetime": "November8"}
	Dar- NO_QUERY ius miles在11月8日的比賽中投中了多少次跳躍投籃		NO_QUERY
	Darius Miles 在2000年11月8日的比賽中投中了多少次跳投 2000年11月8日的比賽	Darius Miles跳投在2000年11月8日投中了 跳投	{"domain": "sports", "sport_type": "其他", "person": "達拉斯·邁爾斯", "datetime": "2000年11月8日"}

Table 9 Last-token Latency breakdown, showing median (P50) and 90th percentile (P90) timings, for the Qwen2.5-7B in Open Book Setting on AudioCRAG-Synthetic (First Token Latency=5.9 sec in T. 1).

Model	Token	P	Latency (sec)			
			Tool Latency		Response Gen	Total
			Query Gen	Tool Results Gen		
Qwen2.5-7B	Last Token	P50	0.59	2.78	16.70	20.07
		P90	0.85	4.90	42.41	48.16

Table 10 Post-training Parameters of OpusLM

Parameter	**Value**
train_micro_batch_size_per_gpu	1
gradient_accumulation_steps	2
epochs	2
gradient_clipping	1.0
bf16 enabled	true
optimizer type	Adam
optimizer lr	0.00001
optimizer betas	[0.9, 0.95]
optimizer eps	1e-8
optimizer weight_decay	3e-7
optimizer adam_w_mode	true
scheduler type	WarmupDecayLR
scheduler warmup_type	linear
scheduler total_num_steps	21534
scheduler warmup_num_steps	1077
scheduler warmup_min_lr	0
scheduler warmup_max_lr	0.00001

Table 11 Post-training Parameters of Qwen-OMNI Thinker

Parameter	**Value**
bf16	True
gradient_accumulation_steps	4
epochs	1
gradient_clipping	1.0
learning_rate	7e-6
lr_scheduler_type	cosine
warmup_ratio	0.05
per_device_train_batch_size	1
weight_decay	0.01

表9 最後一個 token 延遲分解，顯示中位數（P50）和90百分位數（P90）的時間，針對Qwen2.5-7B在開放書籍設定下於AudioCRAG-Synthetic（第一個 Token 延遲=5.9秒在T. 1）。

模型	Token	P	延遲（秒）			
			工具延遲		回應生成	總計
			查詢生成	工具結果生成		
Qwen2.5-7B	最後一個 Token	P50	0.59	2.78	16.70	20.07
		P90	0.85	4.90	42.41	48.16

Table10 OpusLM的後訓練參數

參數	**值**
訓練_微_批次_大小_per_gpu	1
漸層_累積_階段	2
週期	2
漸層_剪裁	1.0
bf16 啟用	true
優化器類型	Adam
優化器 lr	0.00001
優化器 betas	[0.9, 0.95]
優化器 eps	1e-8
優化器權重_衰退	3e-7
優化器 adam_w_模式	true
調度器類型	WarmupDecayLR
調度器 warmup_類型	線性
排程器總_數_步	21534
排程器預熱_數_步	1077
排程器預熱_min_lr	0
排程器預熱_max_lr	0.00001

表11 訓練後參數 Qwen-OMNI Thinker

參數	**值**
bf16	True
漸層_累積_步驟	4
epoches	1
漸層_剪裁	1.0
學習_率	7e-6
lr_調度器_類型	餘弦
暖機_比例	0.05
per_設備_訓練_批次_大小	1
權重_衰退	0.01

Table 12 Post-training Parameters of Qwen-OMNI Talker

Parameter	Value
bf16	True
gradient_accumulation_steps	4
gradient_clipping	1.0
epochs	2
learning_rate	5e-5
per_device_train_batch_size	1
lr_scheduler_type	linear
warmup_ratio	0.0
weight_decay	0.01

表12 訓練後參數 Qwen-OMNI Talker

參數	值
bf16	True
梯度_累積_步數	4
梯度 _裁剪	1.0
世代	2
學習_率	5e-5
per_裝置 _訓練 批次大小_ _	1
學習率_調度器_類型	線性
暖機_比例	0.0
權重_衰退	0.01